



FORTEST

Testing Technology. Together.

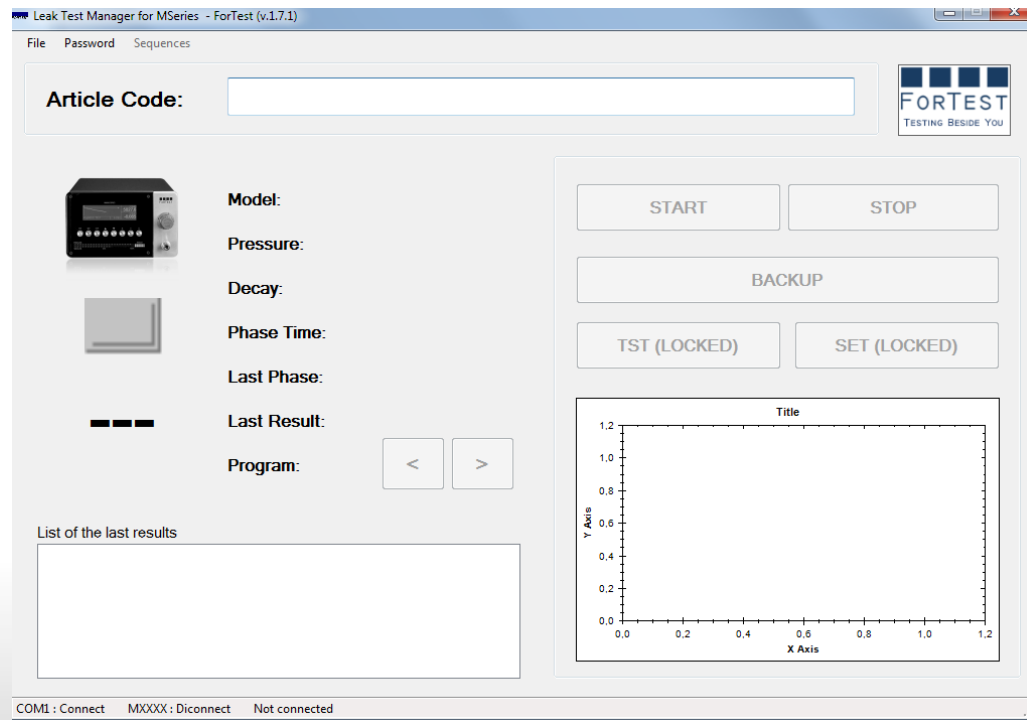


Leak Test Manager for M-Series

PC Software application for manage data and results of ForTest M-Series equipment

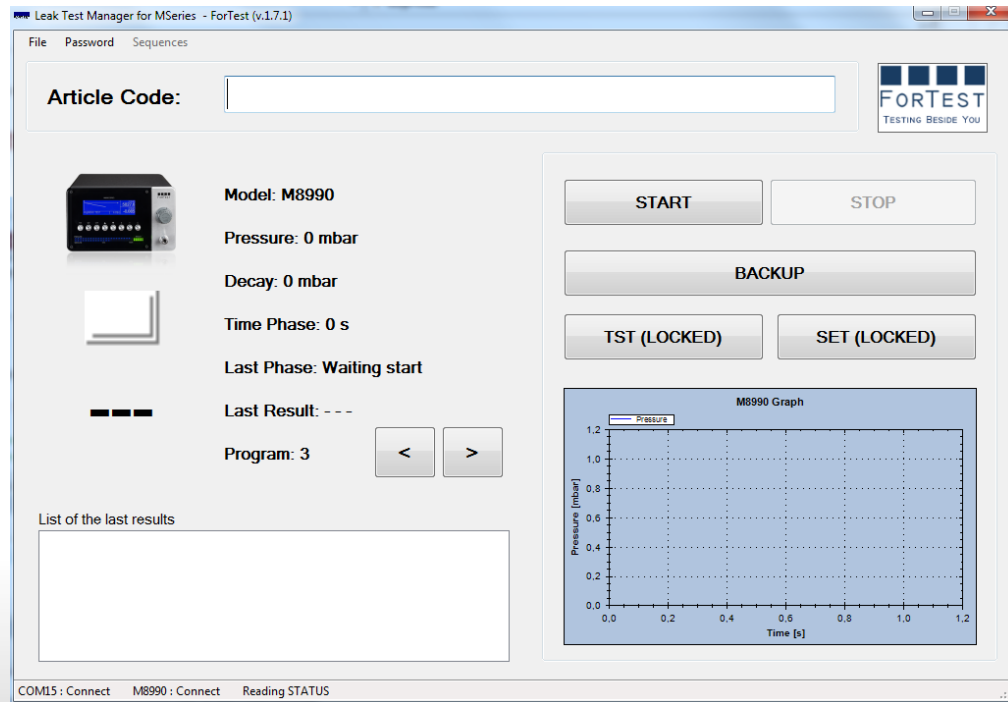
MAIN SCREEN

Device is disconnected (all the controls in grey are disabled)



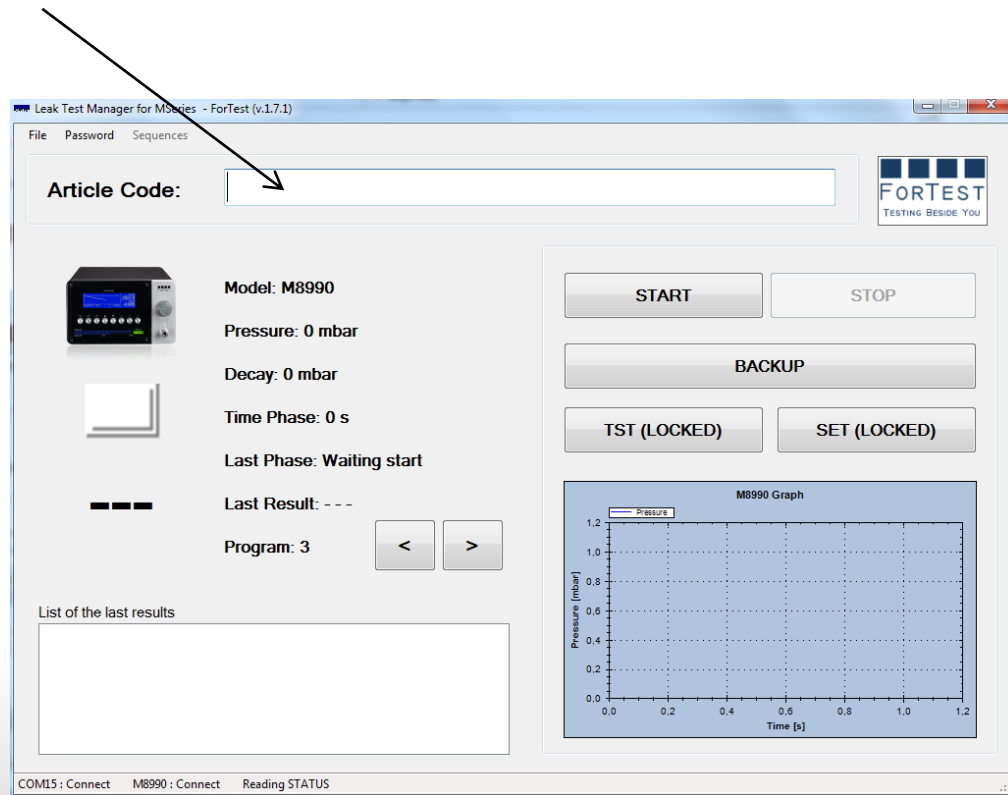
MAIN SCREEN

Device is connected (all the controls are enabled)



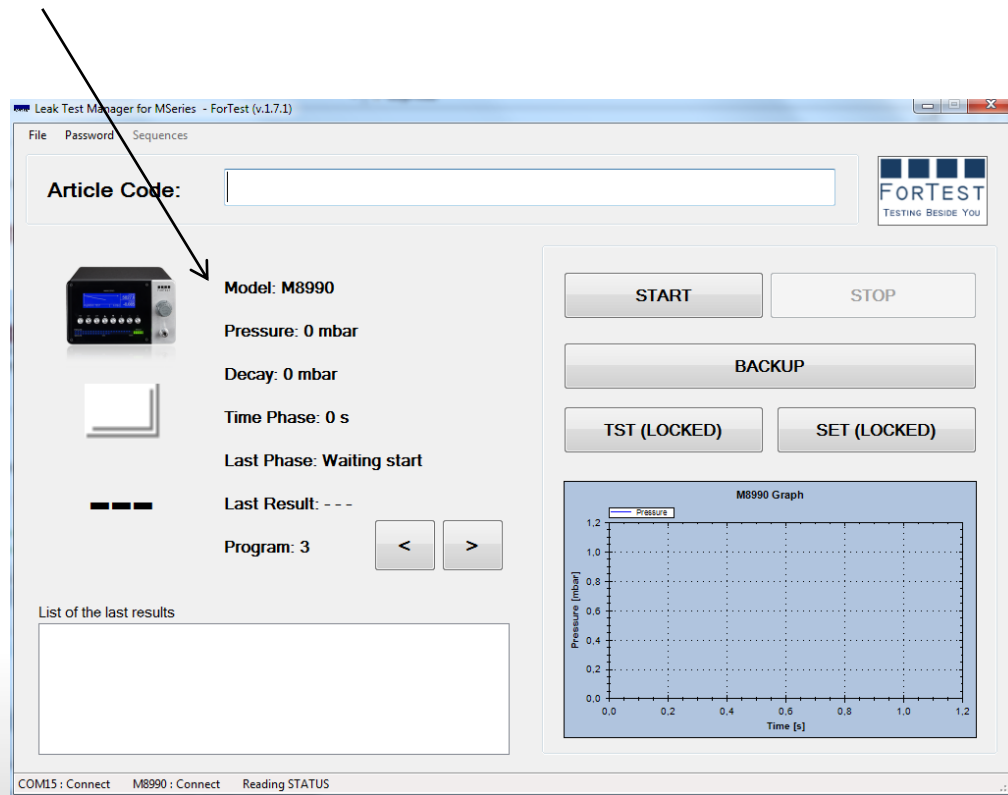
MAIN SCREEN

Article code: is the alphanumerical code that will be recorded into the test archive. It can be insert manually or via a barcode scanner (see options)



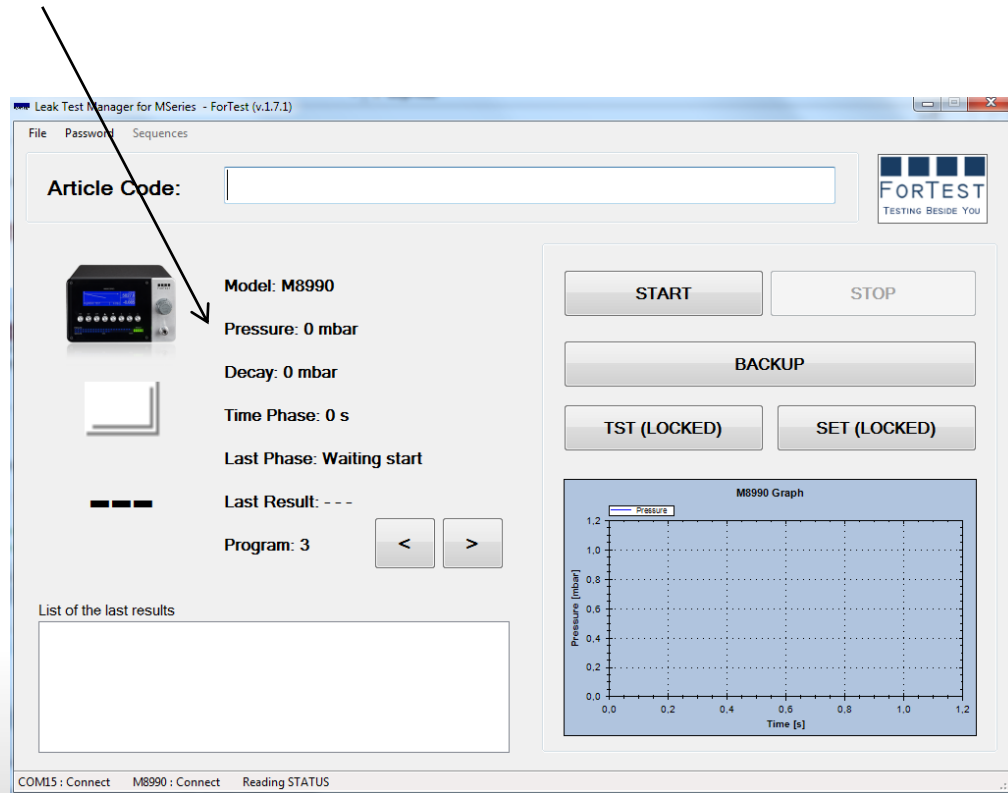
MAIN SCREEN

Model: is the model of the equipment connected.



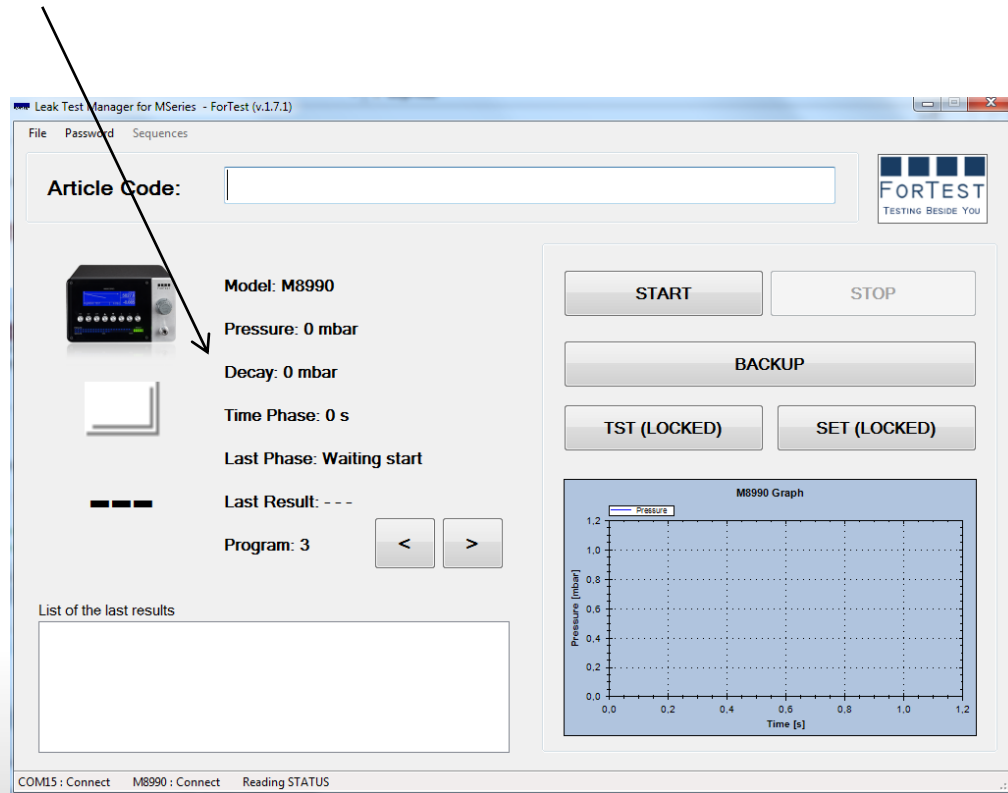
MAIN SCREEN

Pressure: is current pressure evaluated by the device



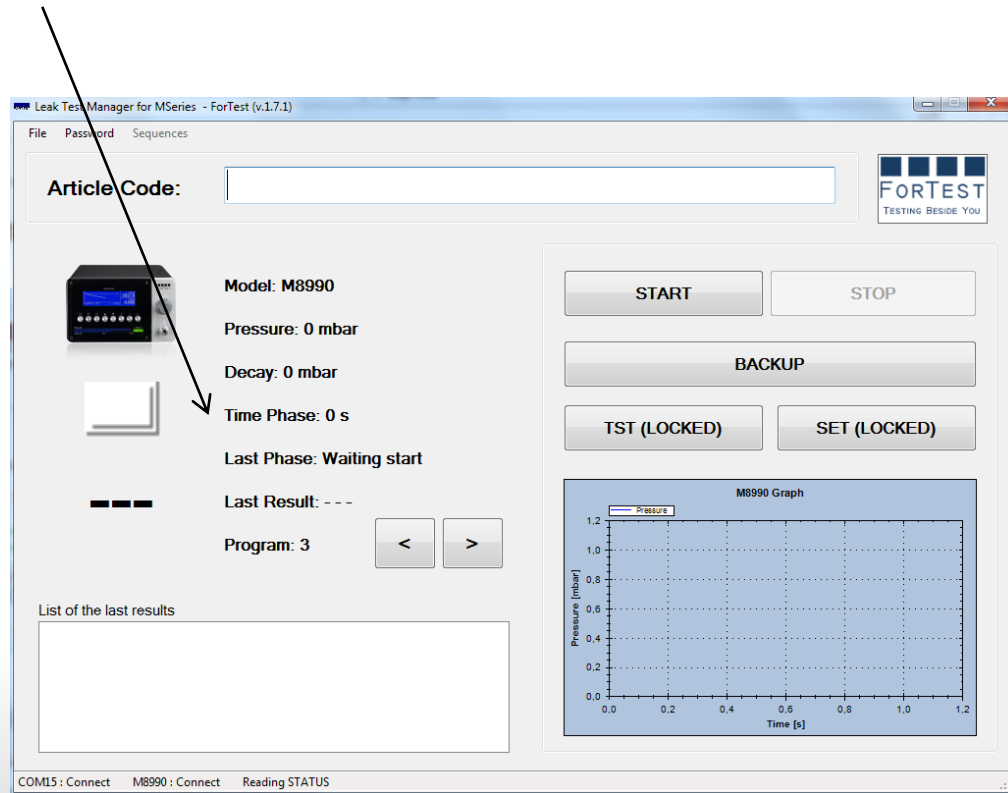
MAIN SCREEN

Decay: is current decay evaluated on the device



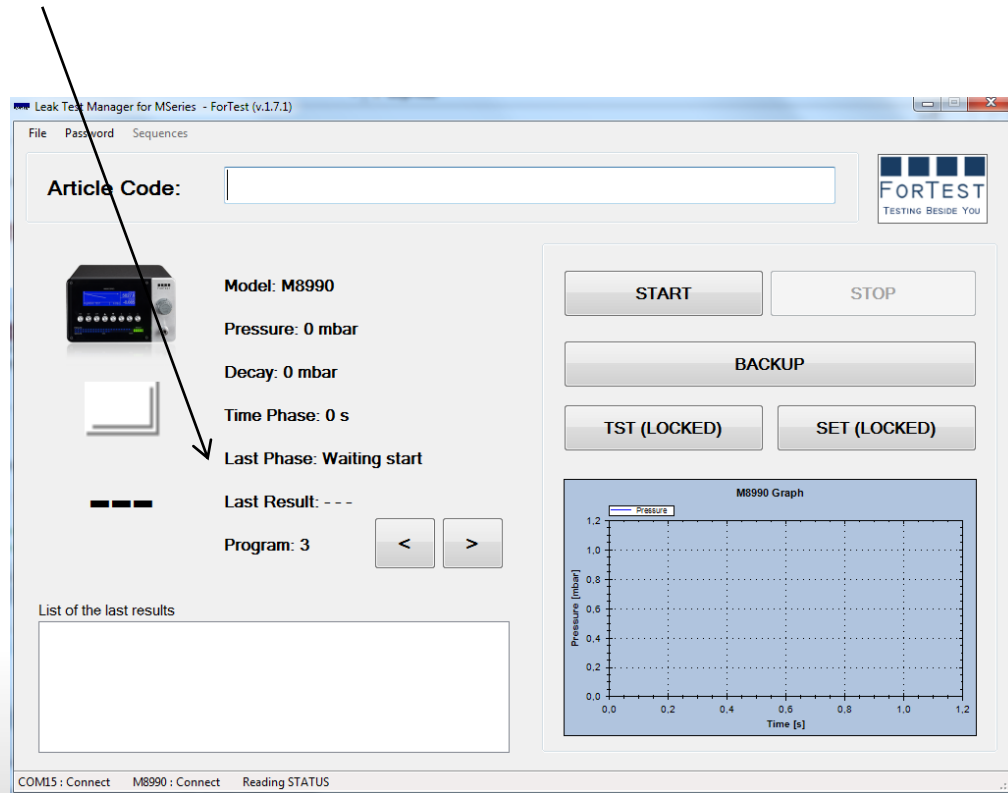
MAIN SCREEN

Time phase: is the time in seconds remaining to the next phase



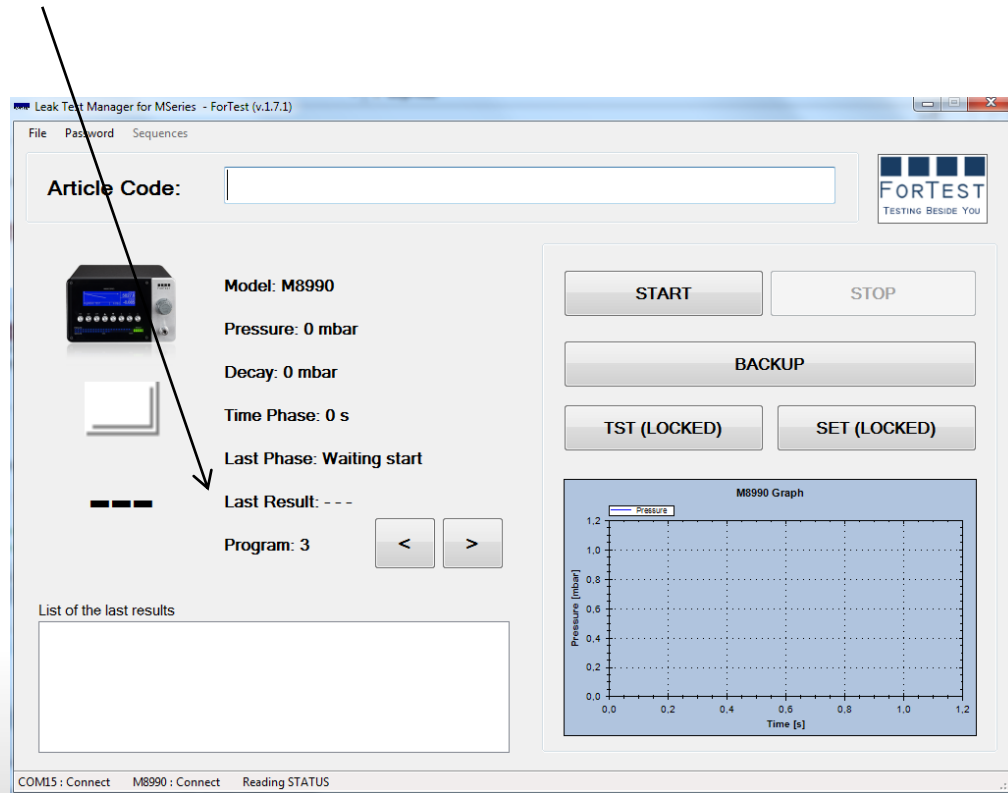
MAIN SCREEN

Last phase: is the last phase evaluated during the full cycle of test (for example : “Waiting start”, “Filling”, “Settling” , “Test”)



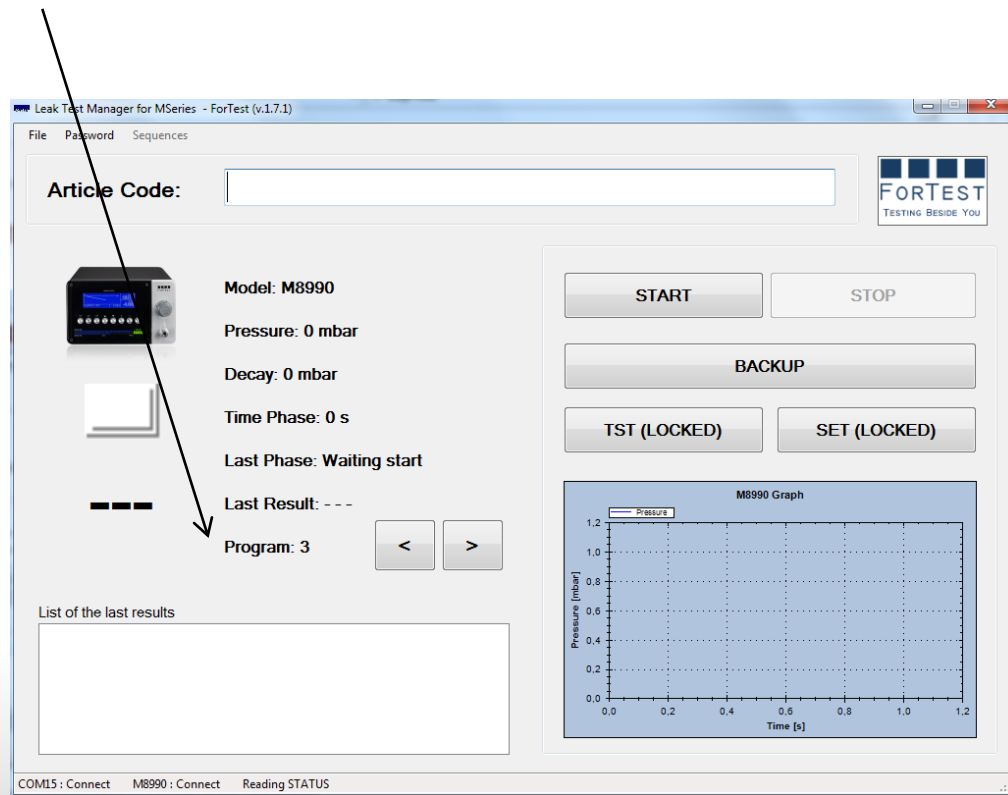
MAIN SCREEN

Last result: is the last result evaluated at the end of test.
For example : “Good” , “Abort” , “Bad”



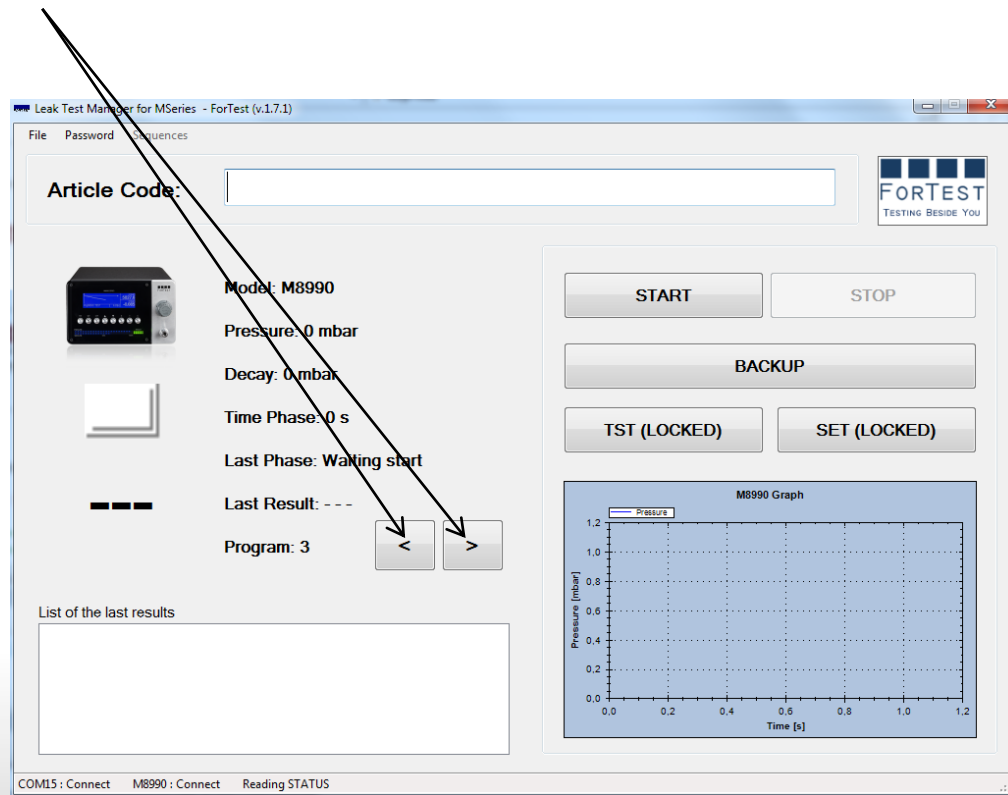
MAIN SCREEN

Program number: is the current program of test running on the device



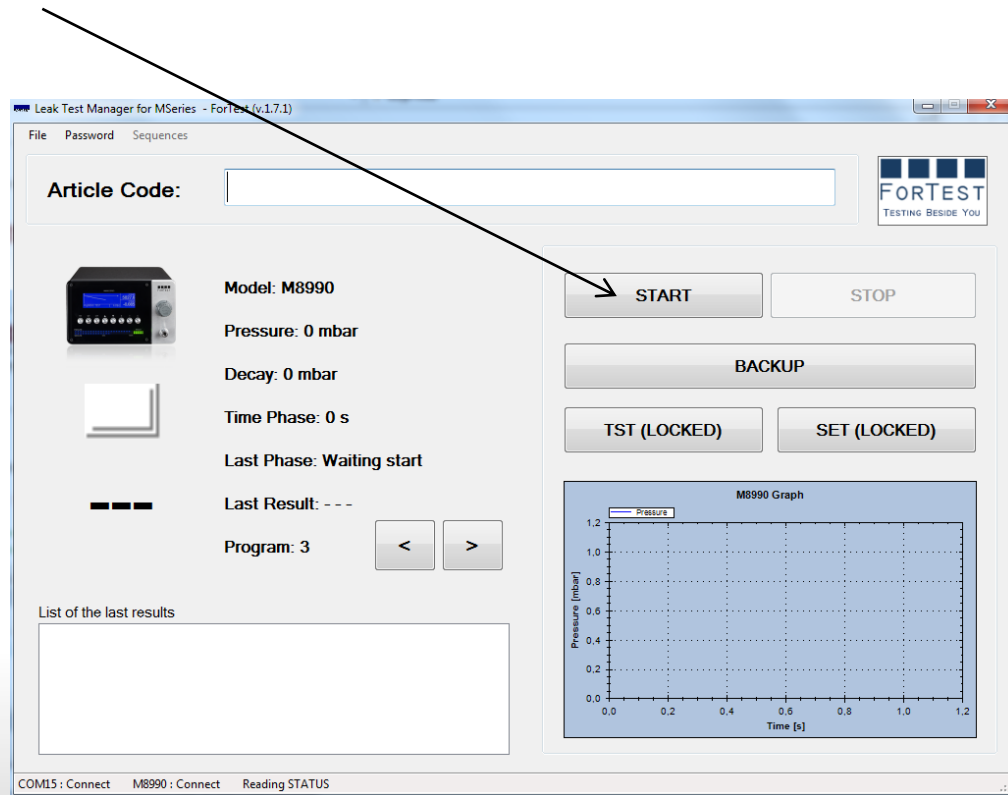
MAIN SCREEN

Program switch: change forward or backward between test programs



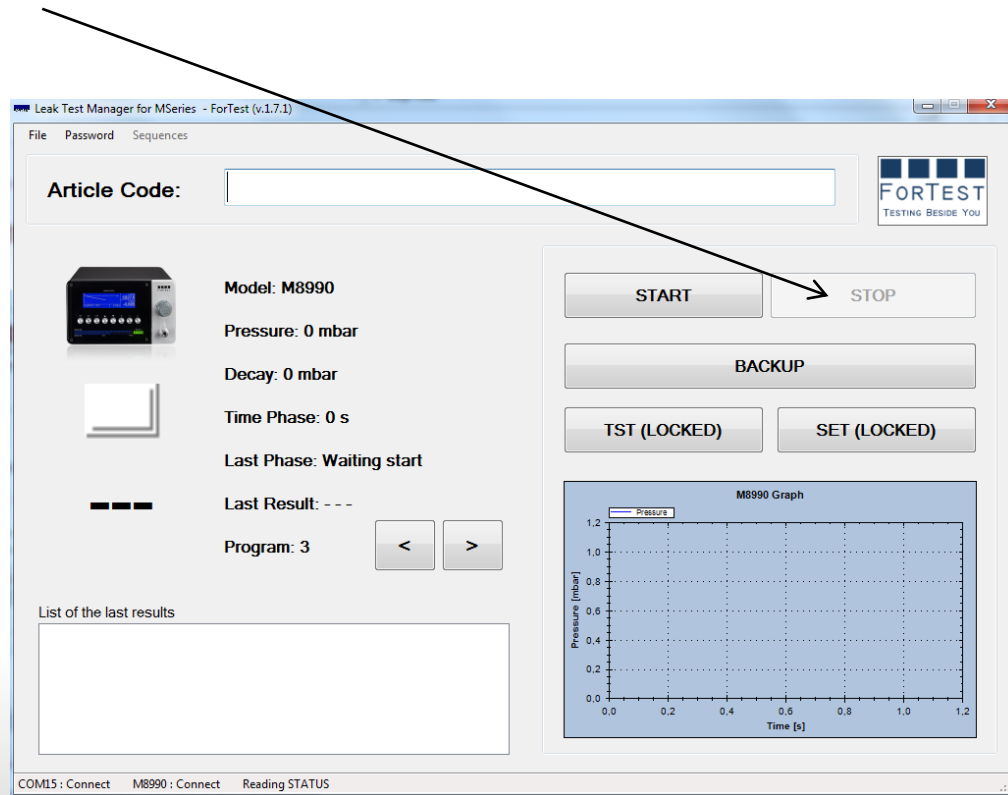
MAIN SCREEN

START: start the current test program



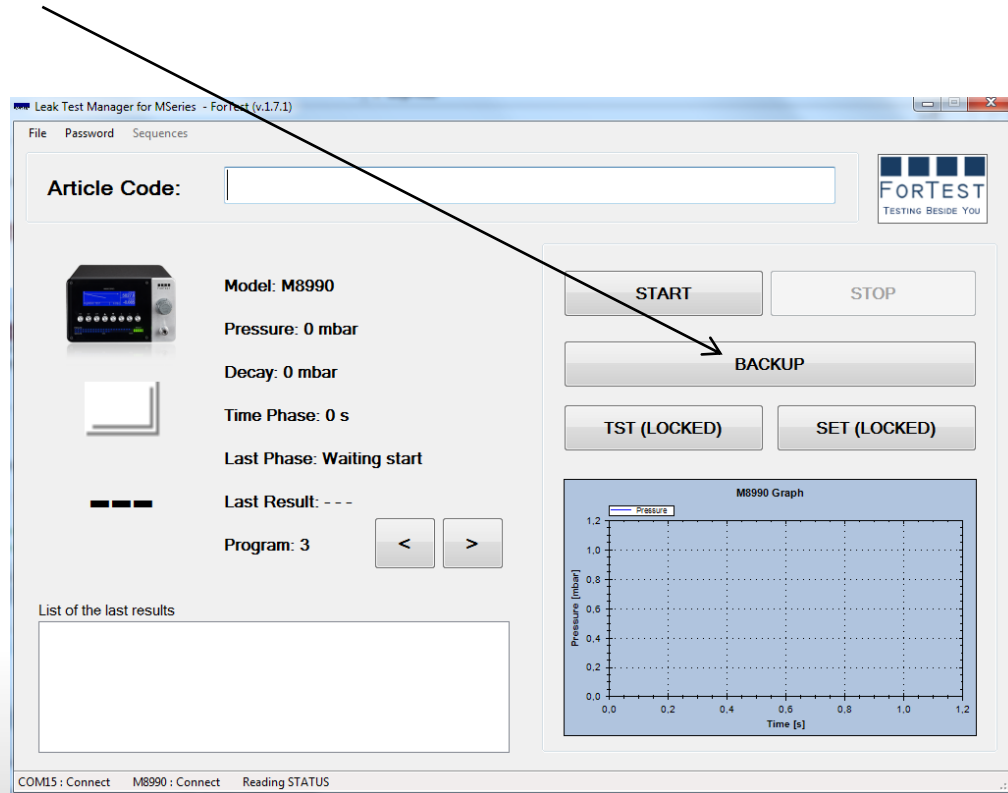
MAIN SCREEN

STOP: stop the current test program



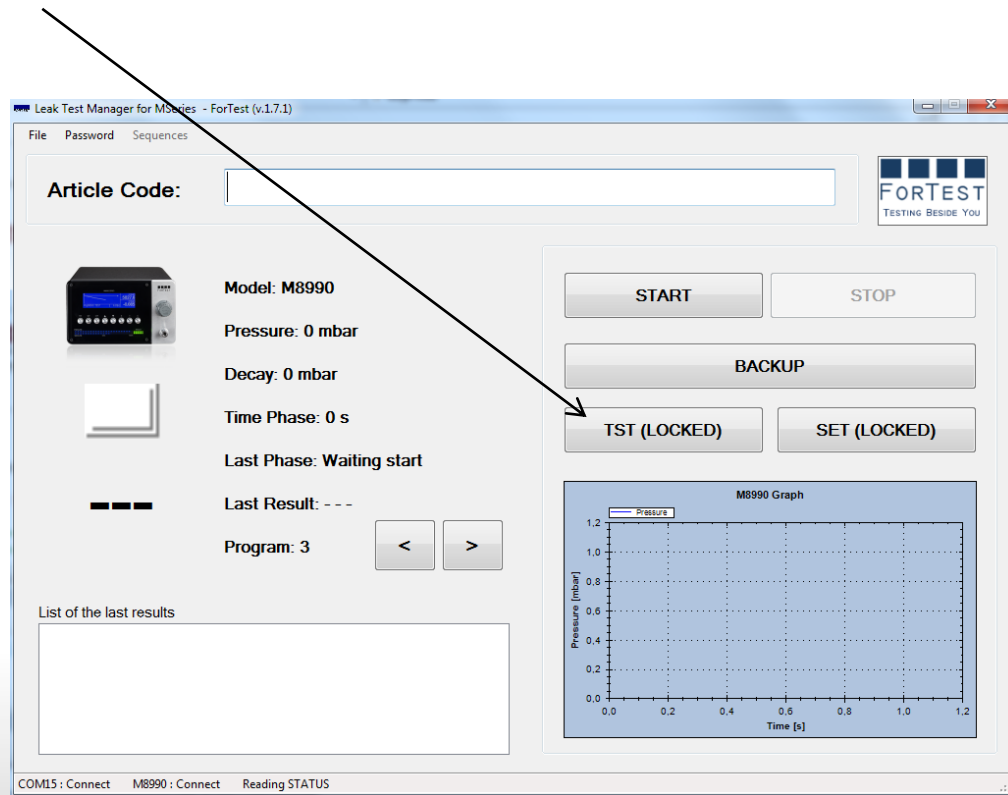
MAIN SCREEN

BACKUP: perform a copy of all programs starting from program 0 to a Determinated program into a XML files. See “BACKUP” for more informations



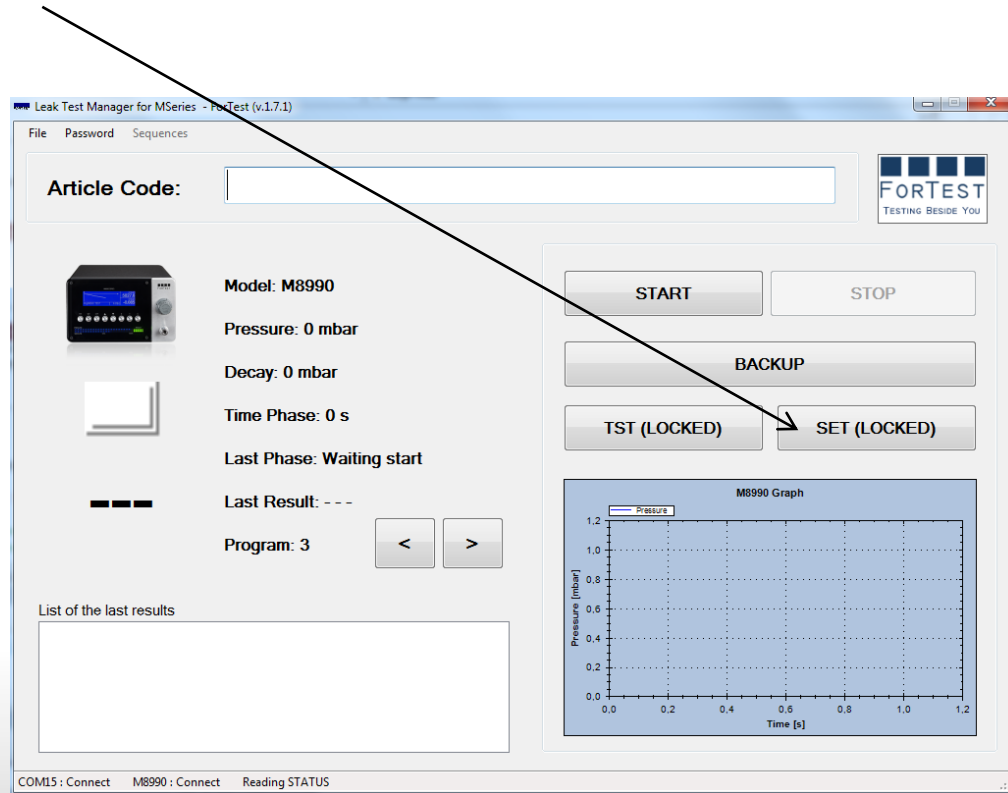
MAIN SCREEN

TST: Modify the testing parameters of a program (See “TST” for more informations) If “LOCKED” it means that is password protected.



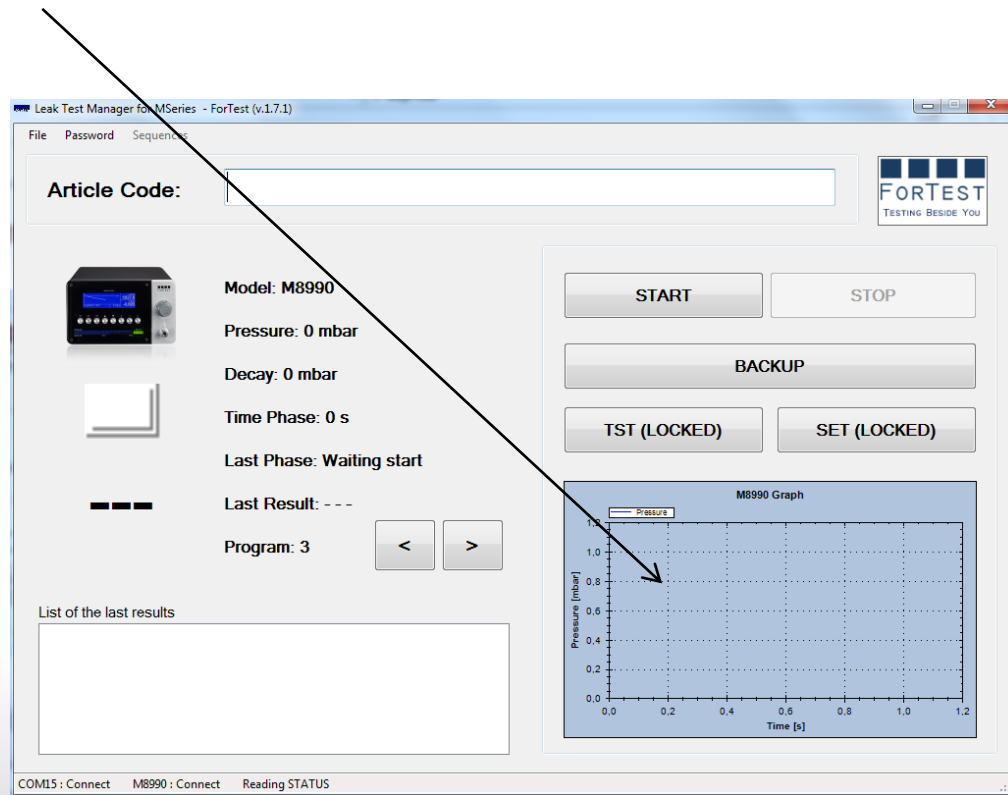
MAIN SCREEN

SET: Modify the setting parameters of the device
If “LOCKED” it means that is password protected.



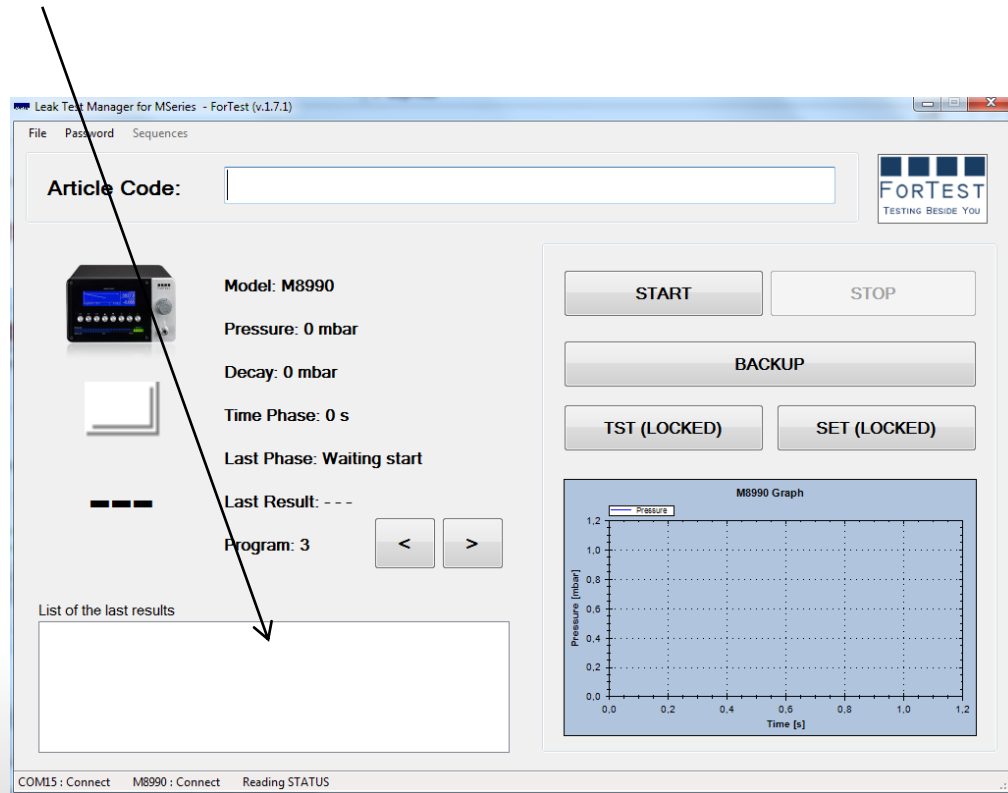
MAIN SCREEN

GRAPH: Graph the pressure and the decay during the test



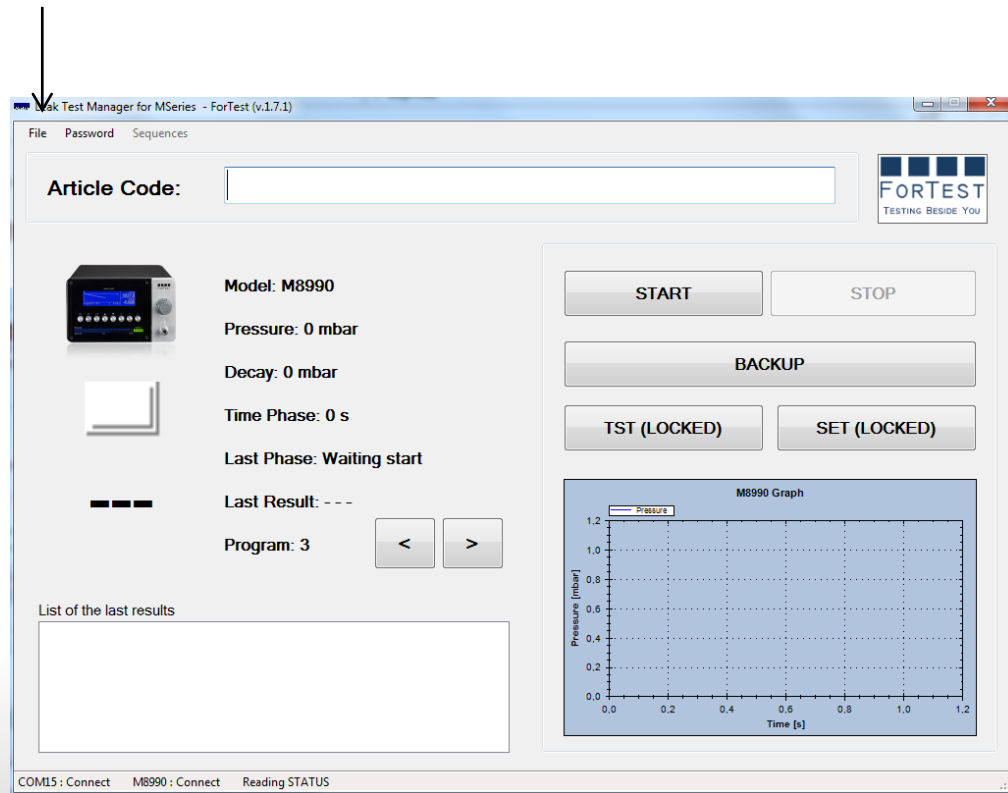
MAIN SCREEN

LAST RESULTS: Show a list of the last results evaluated



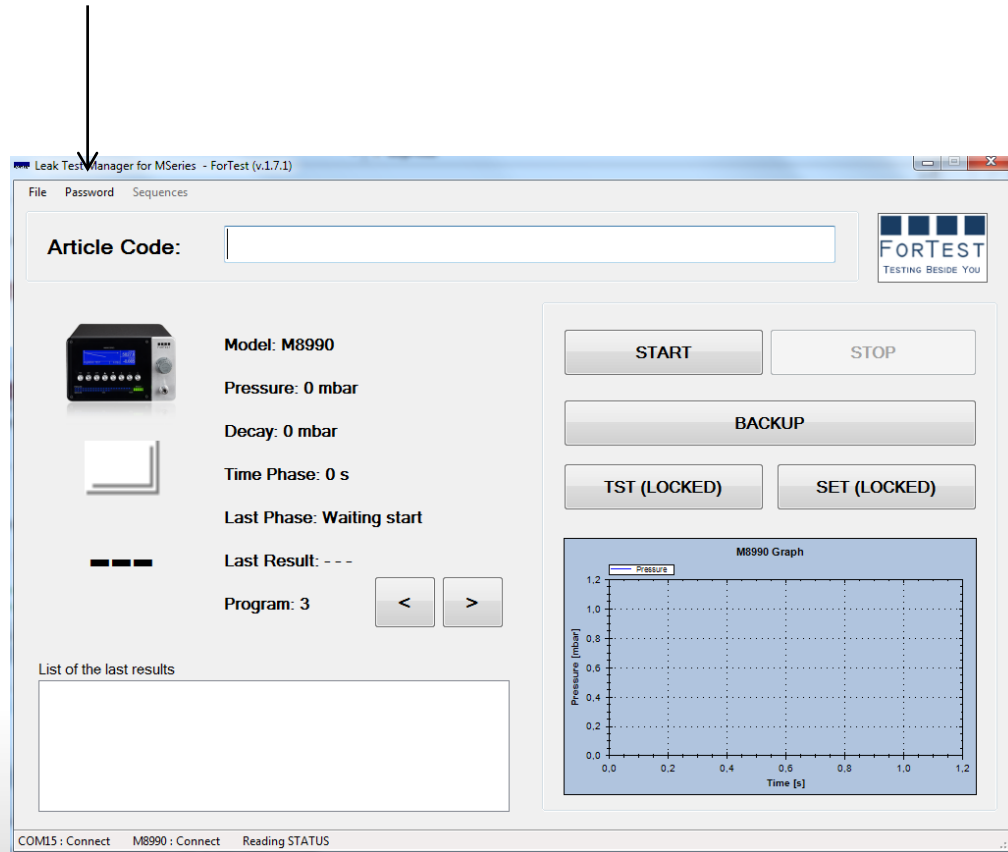
MAIN SCREEN

File: It contain the “OPTION” menu for configure the software



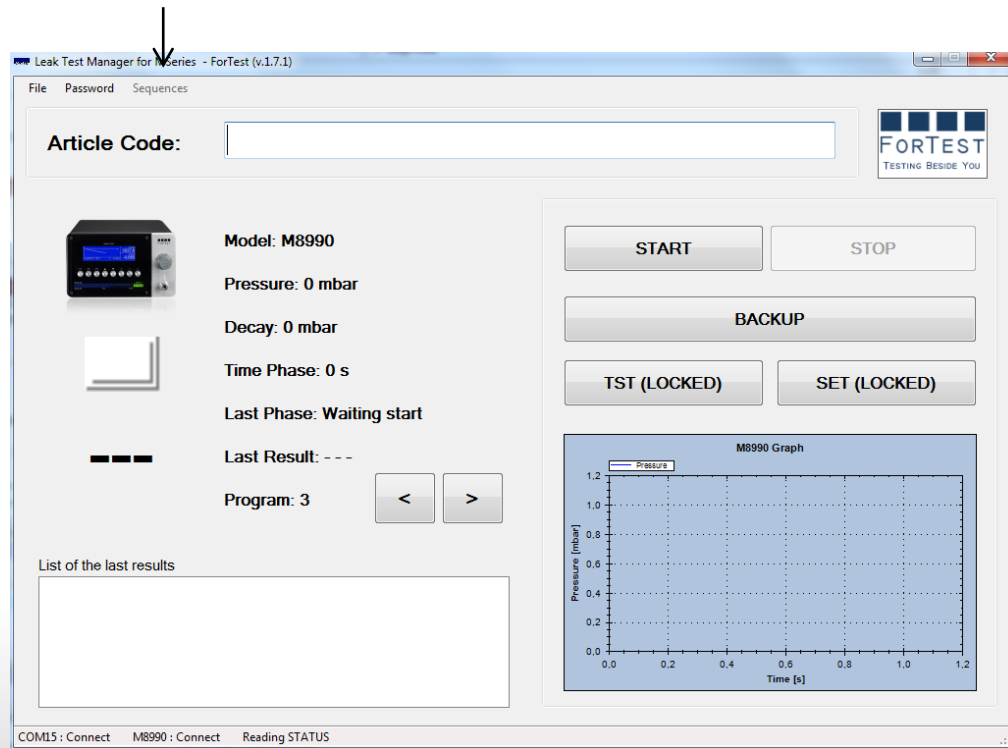
MAIN SCREEN

Password: Lock/unlock controls by password and redefine a new password



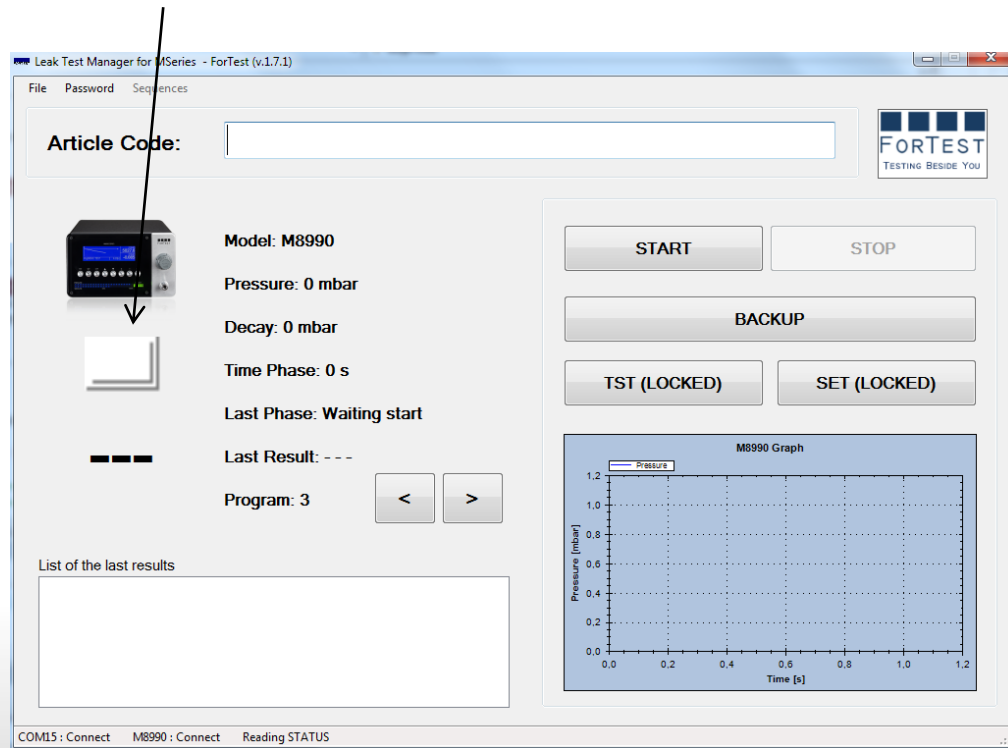
MAIN SCREEN

Sequences: create an association between an article code and the execution of automatic test of sequences. For example is it possible to read a barcode and automatically perform a test on the program 1, if result OK program 2, etc..



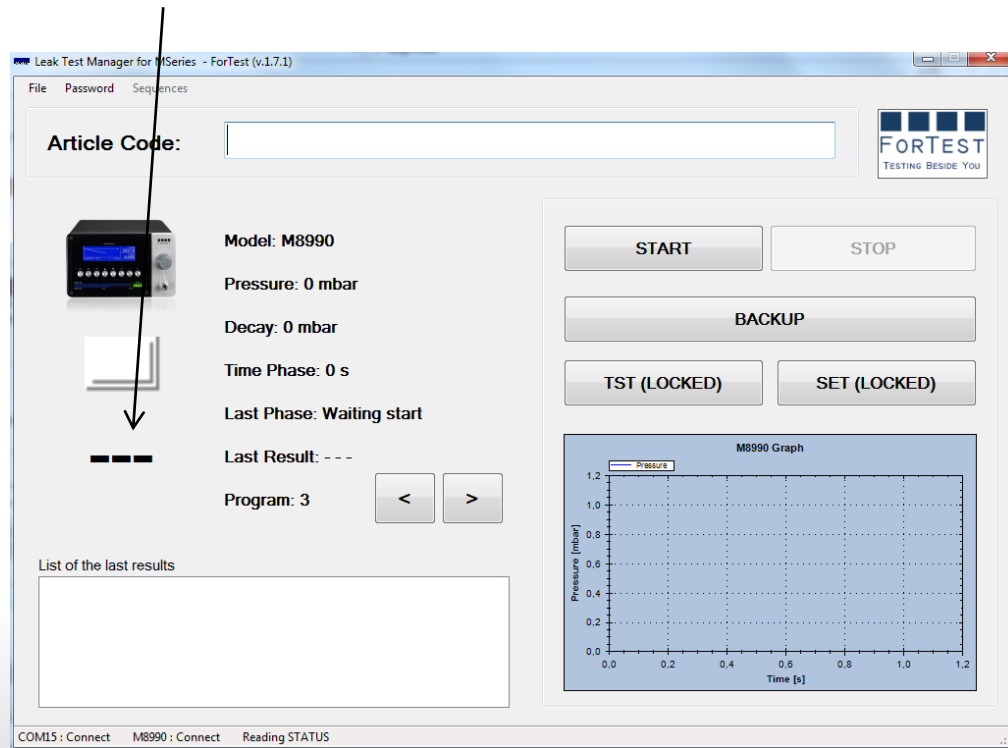
MAIN SCREEN

Semaphore light : GREEN if result is OK, YELLOW if the test is in progress, RED if the test is KO , WHITE if the result is ABORT or not determined



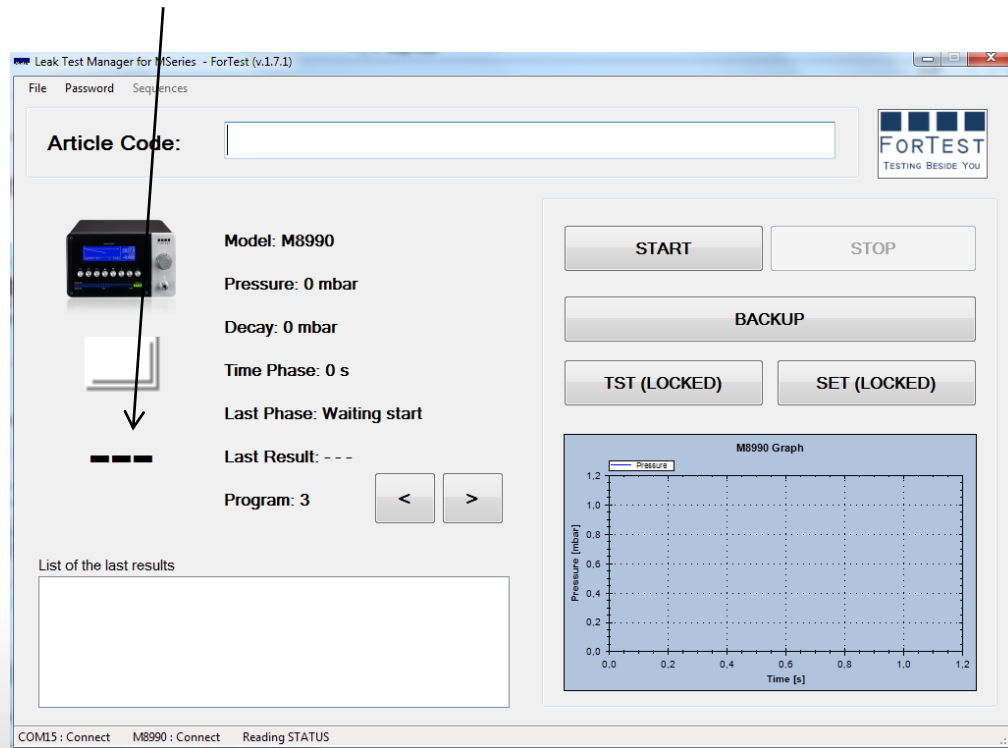
MAIN SCREEN

Short result : It show “OK” for Good result, “KO” for Bad result or “---” if the result is Abort or not determinated



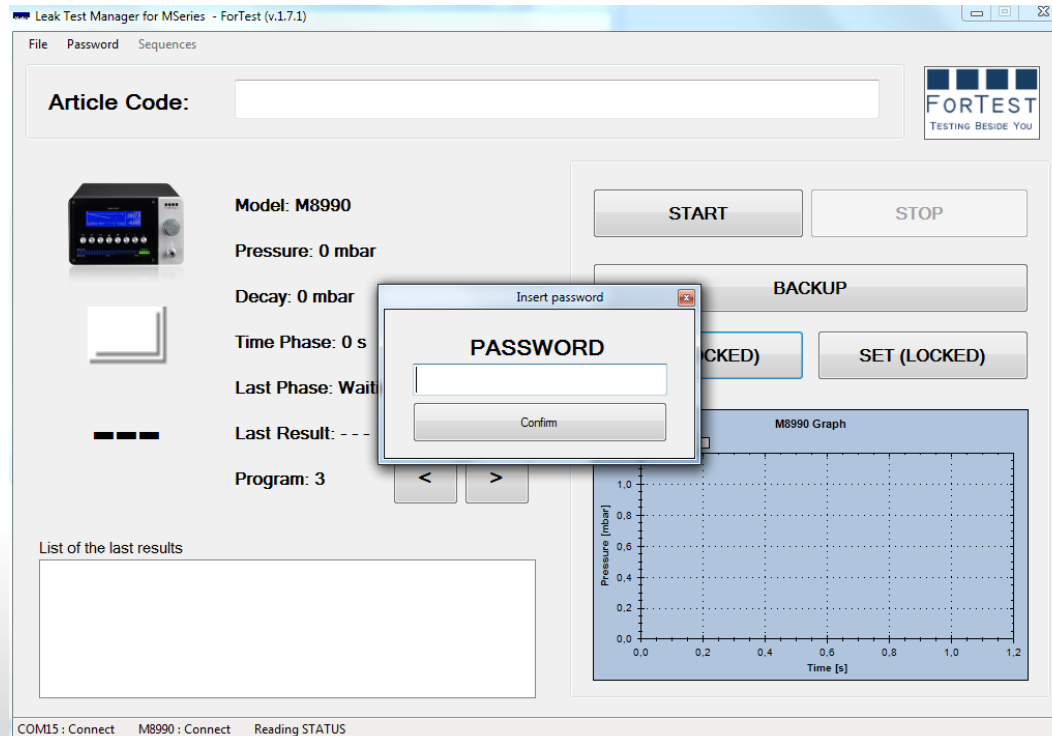
MAIN SCREEN

Short result : It show “OK” for Good result, “KO” for Bad result or “---” if the result is Abort or not determinated



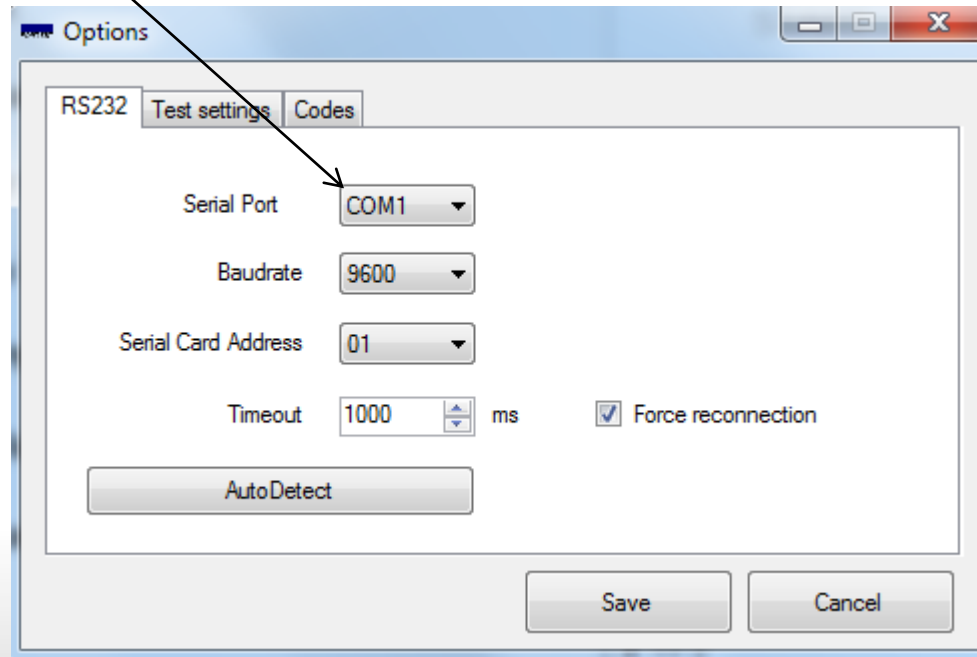
MAIN SCREE

All “**LOCKED**” controls need to be “**UNLOCKED**” with the supervisor password. Default password is “fortest”



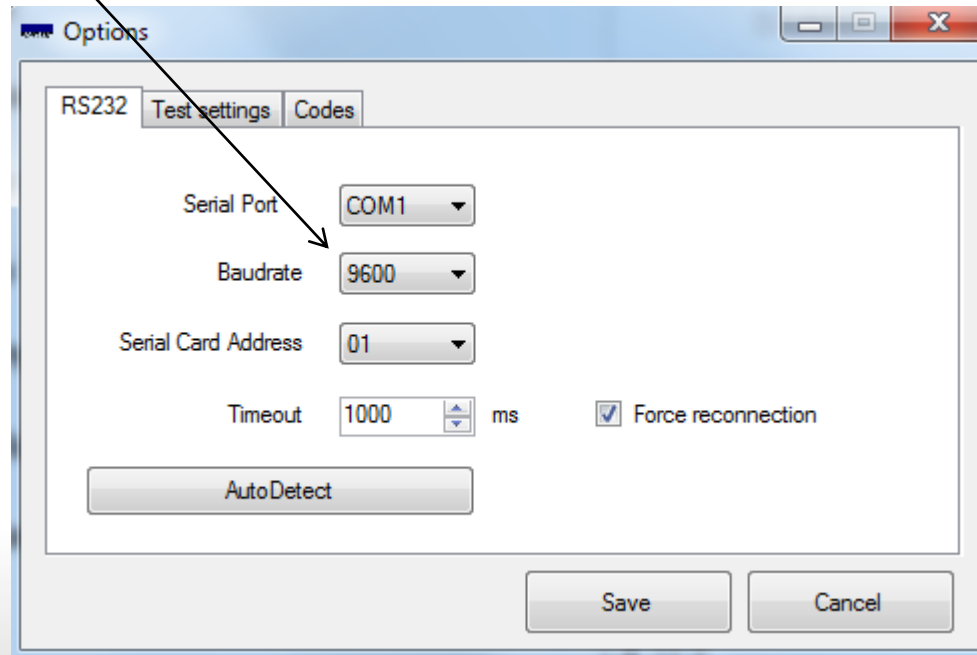
OPTION (RS232)

Serial port : is the serial port name connected to the device



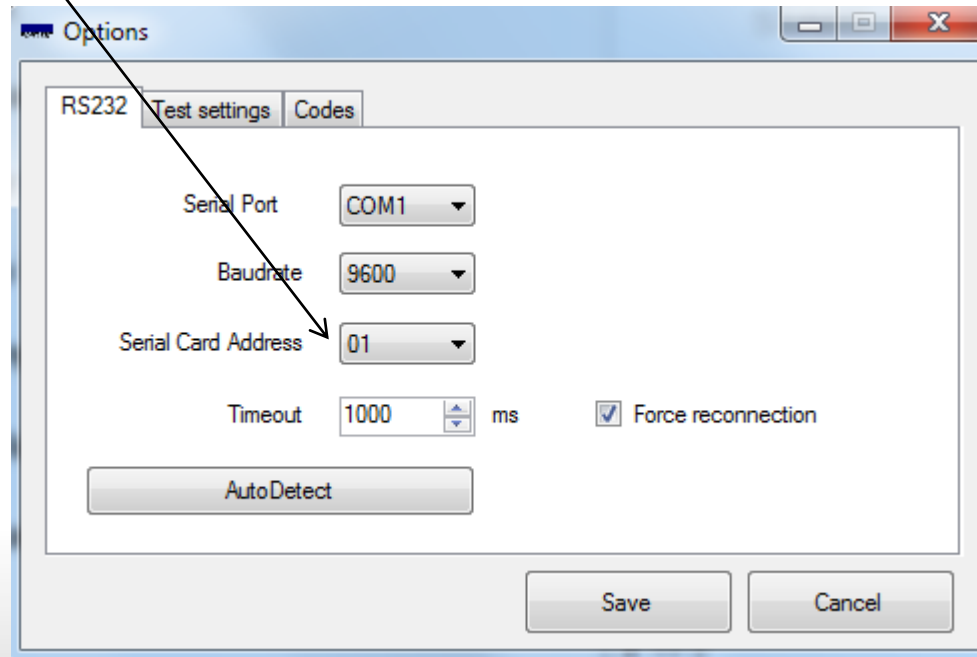
OPTION (RS232)

Baudrate: is the current baudrate of the device



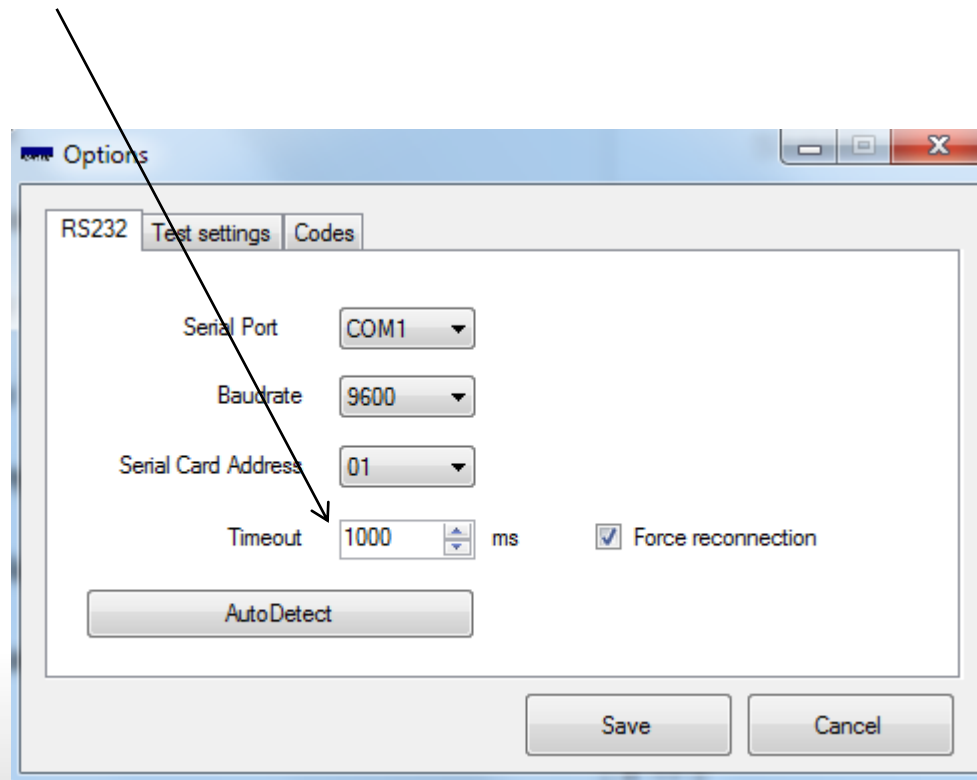
OPTION (RS232)

Serial Card Address: is the address of the device



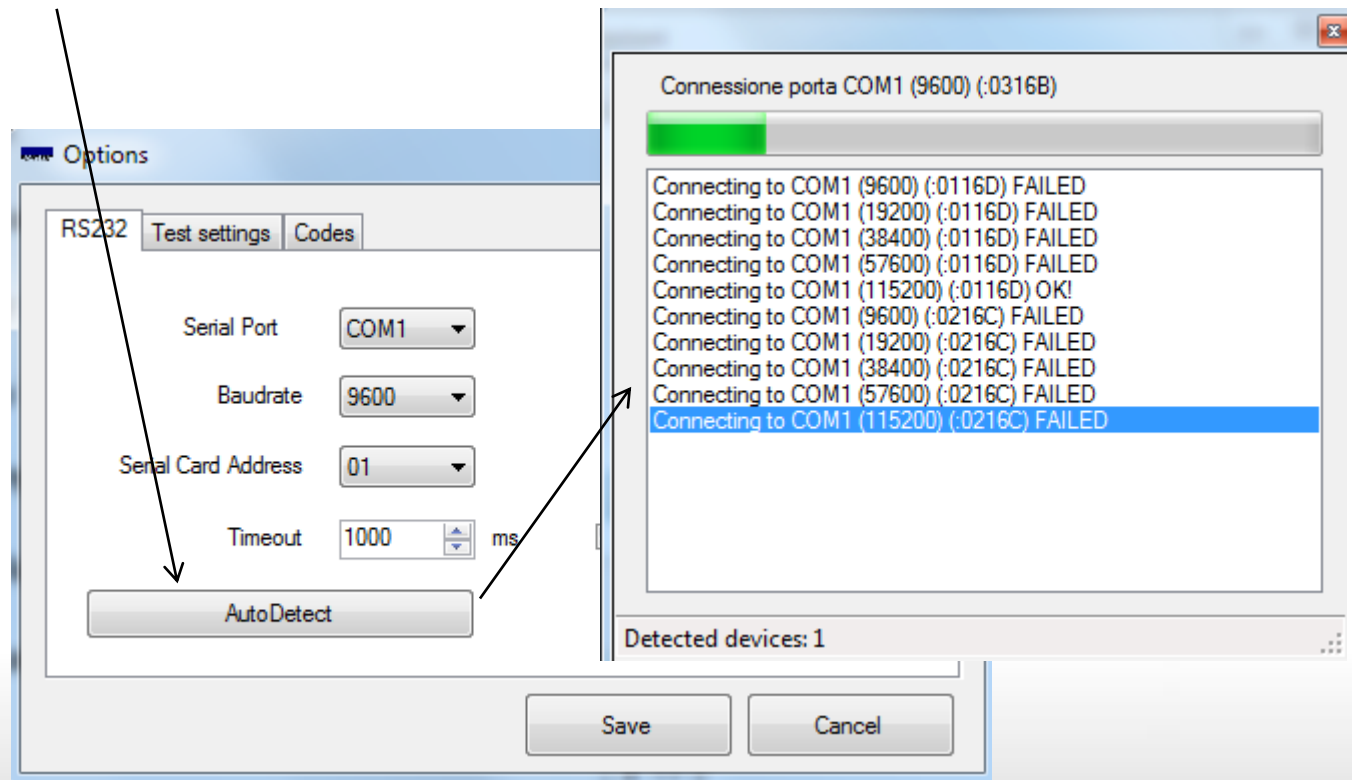
OPTION (RS232)

Timeout: If the software don't receive an RX answer in N-milliseconds it show a timeout error and the communication with the device is lost.



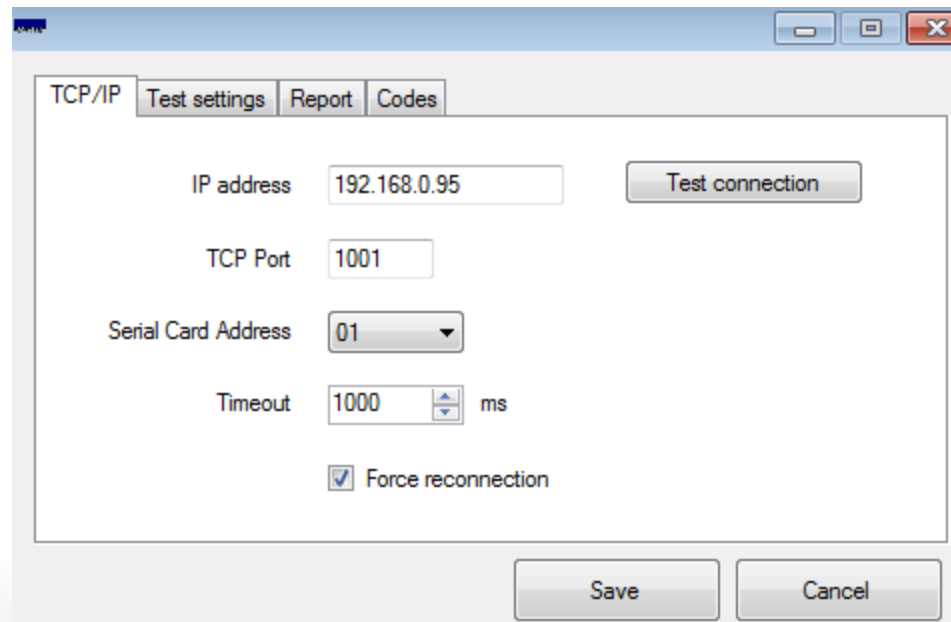
OPTION (RS232)

Autodetect: Perform a scan of the equipment on every serial port and if detected it automatically configure the connection with it



OPTION (TCP/IP)

IP address: IPV4 address of the equipment. Please read the document “Ethernet configuration” before try to connect.



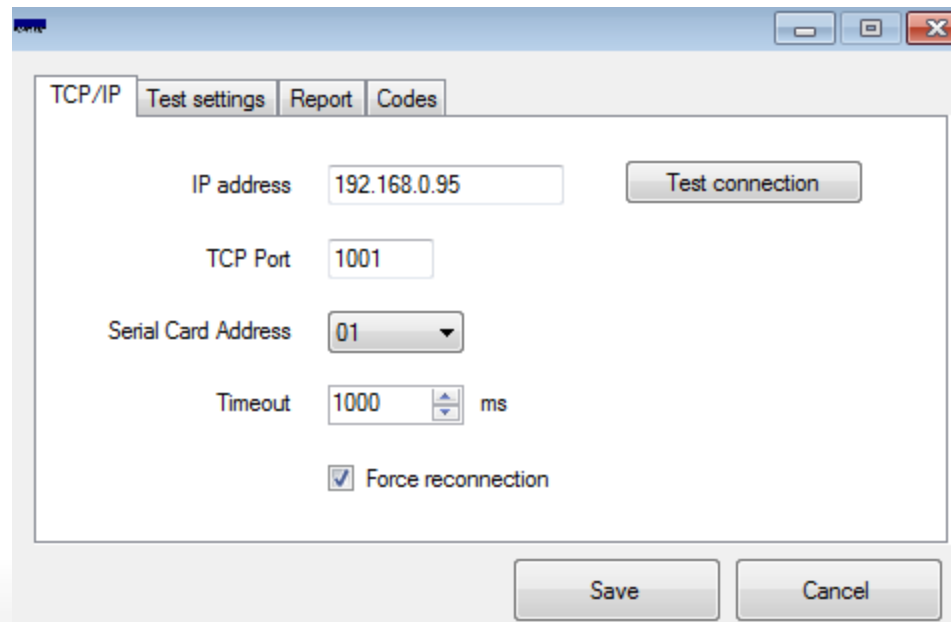
The image shows a software window titled "TCP/IP" with a standard Windows-style title bar (minimize, maximize, close buttons). Inside the window, there are four tabs: "TCP/IP", "Test settings", "Report", and "Codes". The "TCP/IP" tab is currently selected. Below the tabs, there are several input fields and a button:

- IP address:** A text box containing "192.168.0.95".
- TCP Port:** A text box containing "1001".
- Serial Card Address:** A dropdown menu showing "01".
- Timeout:** A spin box containing "1000" followed by "ms".
- Force reconnection:** A checked checkbox.
- Test connection:** A button located to the right of the IP address field.

At the bottom of the window, there are two buttons: "Save" and "Cancel".

OPTION (TCP/IP)

TCP PORT: Tcp port to connect. Default is 1001. If you need to change the port please check the document “Ethernet configuration”



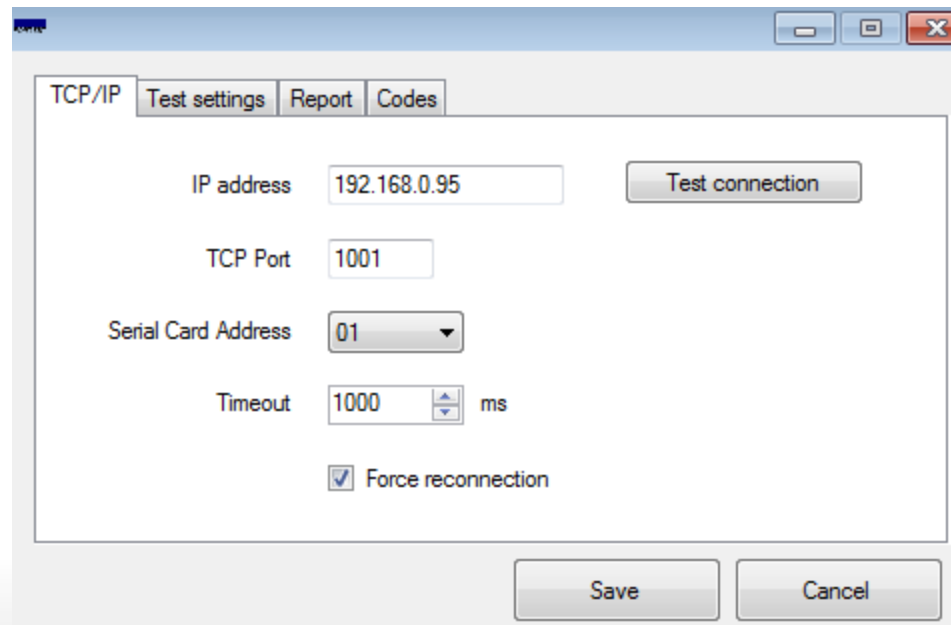
The image shows a software window titled "TCP/IP" with a standard Windows-style title bar (minimize, maximize, close buttons). Inside the window, there are four tabs: "TCP/IP", "Test settings", "Report", and "Codes". The "TCP/IP" tab is currently selected. Below the tabs, there are several input fields and a button:

- IP address:** A text box containing "192.168.0.95".
- TCP Port:** A text box containing "1001".
- Serial Card Address:** A dropdown menu showing "01".
- Timeout:** A spin box showing "1000" with "ms" next to it.
- Force reconnection:** A checked checkbox.
- Test connection:** A button located to the right of the IP address field.

At the bottom of the window, there are two buttons: "Save" and "Cancel".

OPTION (TCP/IP)

Serial Card Address: Default card address is 01 for TCP/IP

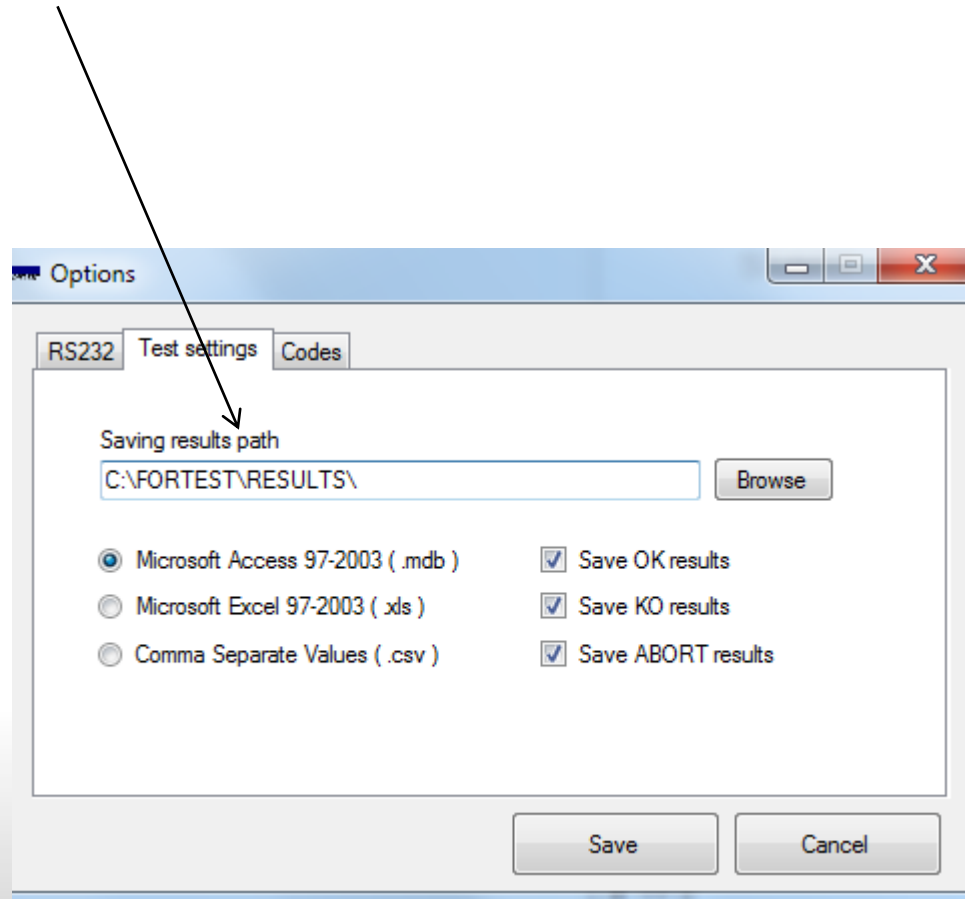


The image shows a software window titled "TCP/IP" with a standard Windows-style title bar (minimize, maximize, close buttons). Inside the window, there are four tabs: "TCP/IP", "Test settings", "Report", and "Codes". The "TCP/IP" tab is currently selected. The main area of the window contains the following fields and controls:

- IP address:** A text input field containing "192.168.0.95".
- TCP Port:** A text input field containing "1001".
- Serial Card Address:** A dropdown menu showing "01".
- Timeout:** A spin box containing "1000" followed by the unit "ms".
- Force reconnection:** A checked checkbox.
- Test connection:** A button located to the right of the IP address field.
- Save:** A button at the bottom right of the window.
- Cancel:** A button at the bottom right of the window, next to the Save button.

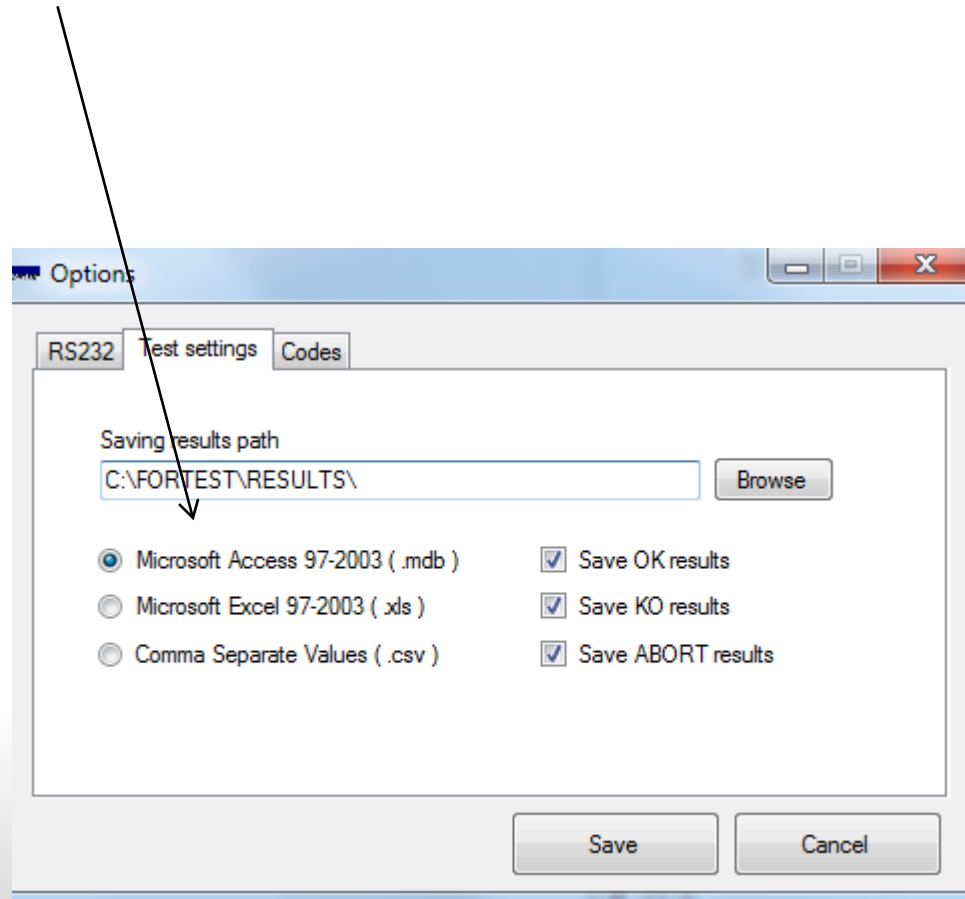
OPTION (TEST)

SAVING RESULT PATH: Is the directory path of the test archive saved



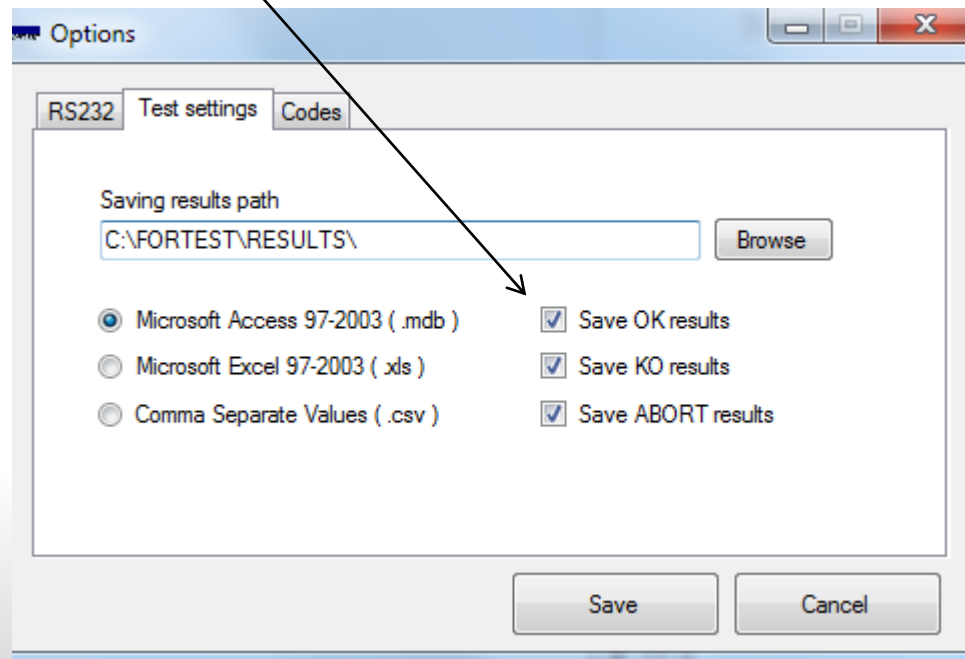
OPTION (TEST)

FORMAT ARCHIVE: Is the format of the archive (MsAccess, Excel , CSV)



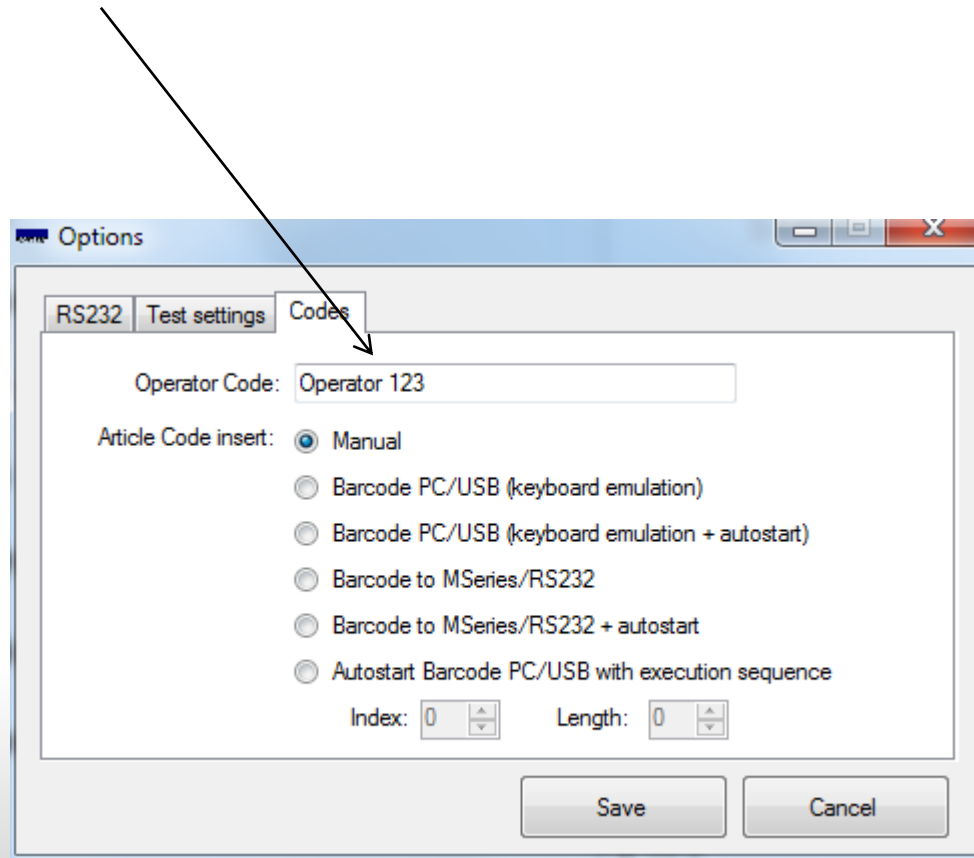
OPTION (TEST)

SAVE RESULTS: Choose to save all “OK”, “KO” or “ABORT” results



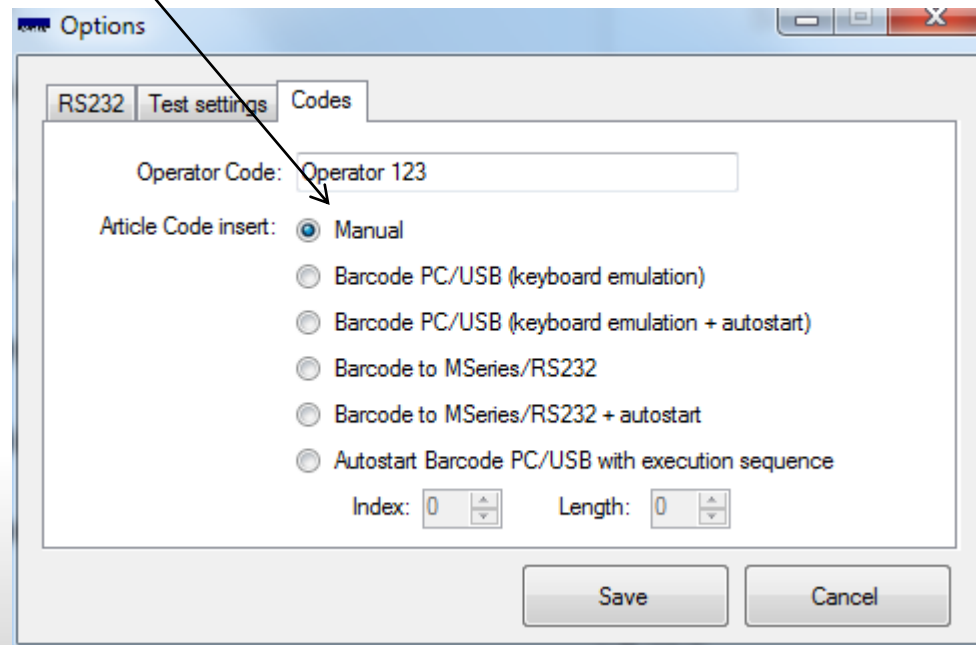
OPTION (BARCODE)

OPERATOR CODE: Is a code that will be inserted into the archive. It is an identification of who made the test.



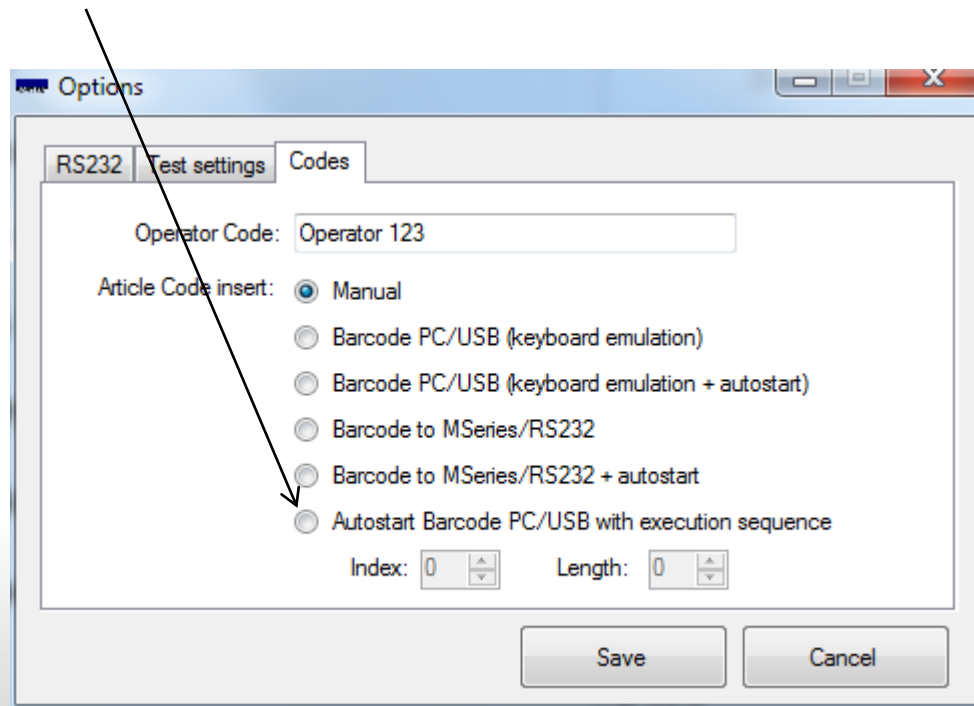
OPTION (BARCODE)

ARTICLE CODE INSERT : specify if the article code is inserted manually from the keyboard, from a barcode connected to PC or connected to the device.



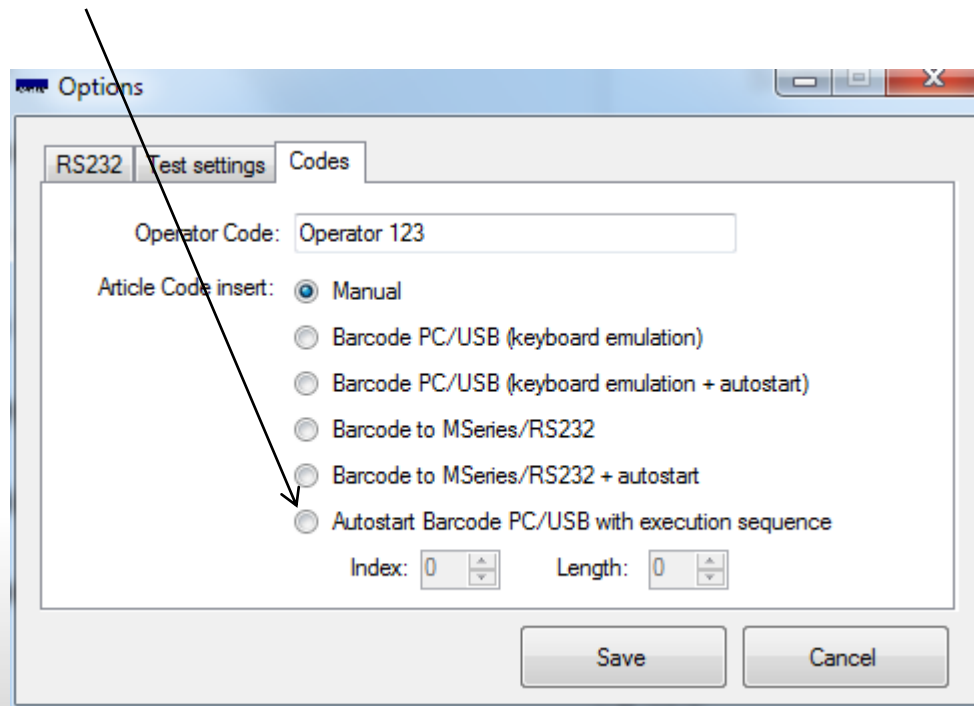
OPTION (BARCODE)

NOTE: “**Autostart barcode PC/USB with execution sequence**” will enable the “SEQUENCES” menu. It permit to auto execute programs when a determinate article is written. **Index** is the starting point from evaluate the code. **Length** is the length of the barcode



OPTION (BARCODE)

For example : If you have the barcode “ABC123DEF456” and you want to filter the barcode to evaluate only to “123DEF” you must use INDEX = 3 , LENGTH = 6 (starting index is 0). Into “SEQUENCES” only the filtered code will be matched



TST

Example of TST parameters of the device. All text parameters must be confirmed with the “ENTER” key pressure

The screenshot displays the ForTest software interface. A central dialog box is open, showing various test parameters for a TST (Time-to-Success) test. The parameters are organized into a table-like structure with labels and input fields. The 'Program' field is set to 3. The 'Kind of test' is set to 'Leak Test'. The 'Filling Pressure' is 5000 mbar, 'Pressure tolerance' is 1 %, 'Filling attempts' is 0, 'Filling phase duration' is 5 s, 'Settling phase duration' is 5 s, 'Test phase duration' is 10 s, 'Discharge phase duration' is 0 s, 'Kind of measurement' is 'Pressure', 'Maximum decay' is 1 mbar, 'Piece's volume' is 0 mbar, and 'Offset on the decay' is 0 mbar. To the right of the dialog box, there are buttons for 'STOP', 'BACKUP', and 'SET (UNLOCKED)'. Below these buttons is a graph titled '990 Graph' showing a pressure decay over time. The x-axis is labeled 'Time [s]' and ranges from 0.0 to 1.2. The y-axis is labeled 'Pressure [mbar]' and ranges from 0.0 to 1.2. The graph shows a blue line representing the pressure decay curve.

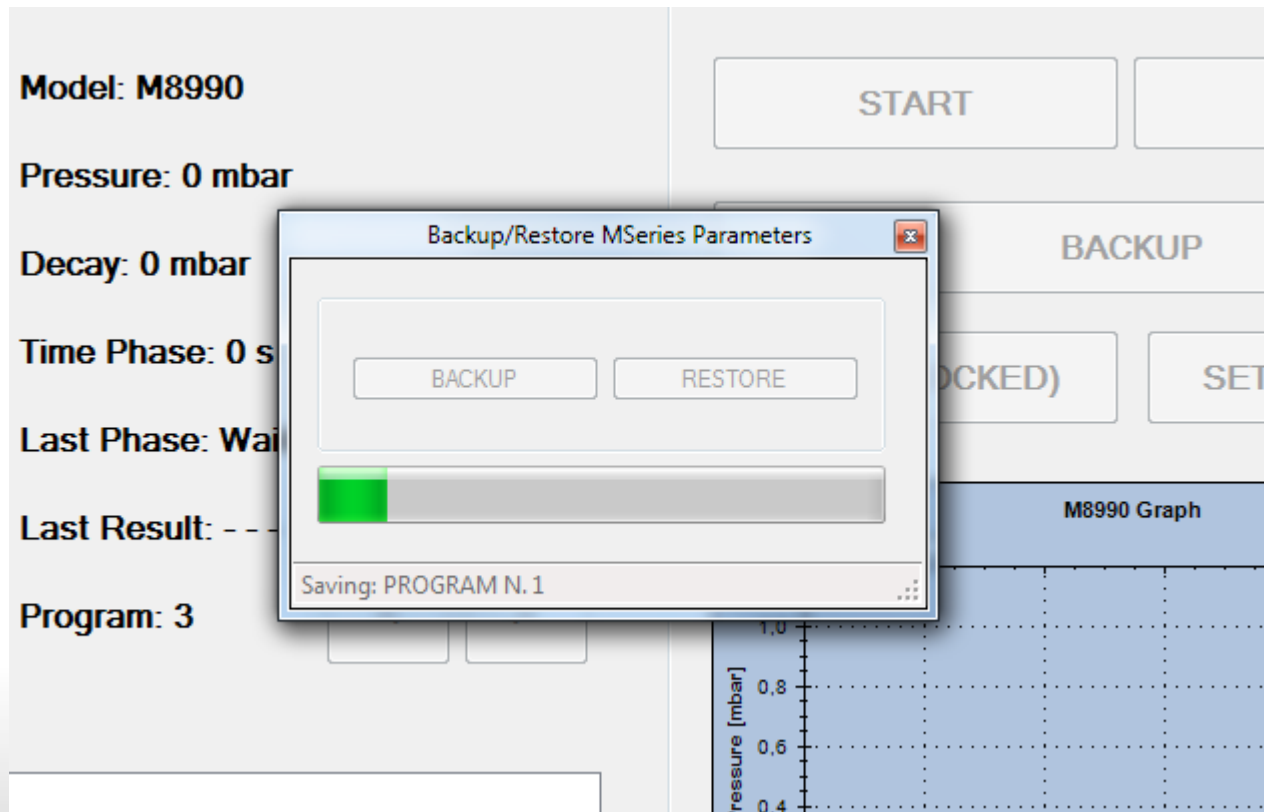
Parameter	Value	Unit
Program	3	
Kind of test	Leak Test	
Filling Pressure	5000	mbar
Pressure tolerance	1	%
Filling attempts	0	
Filling phase duration	5	s
Settling phase duration	5	s
Test phase duration	10	s
Discharge phase duration	0	s
Kind of measurement	Pressure	
Maximum decay	1	mbar
Piece's volume	0	mbar
Offset on the decay	0	mbar

Buttons: STOP, BACKUP, SET (UNLOCKED)

Graph: 990 Graph (Pressure [mbar] vs Time [s])

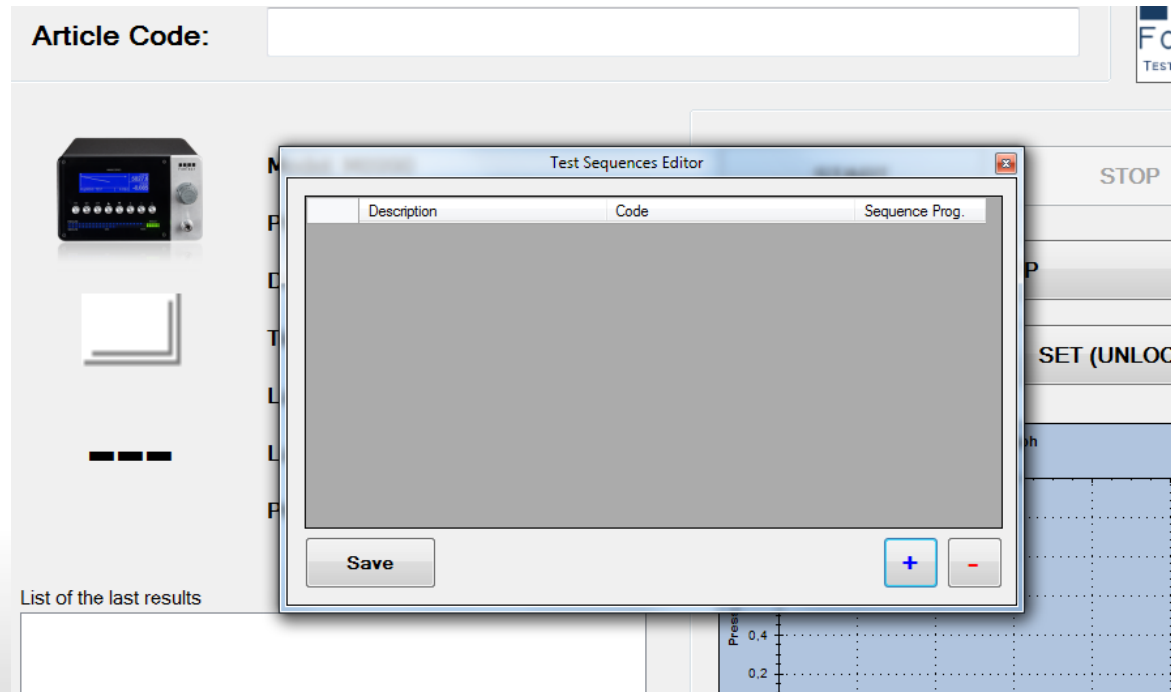
BACKUP

Example of BACKUP of TST programs



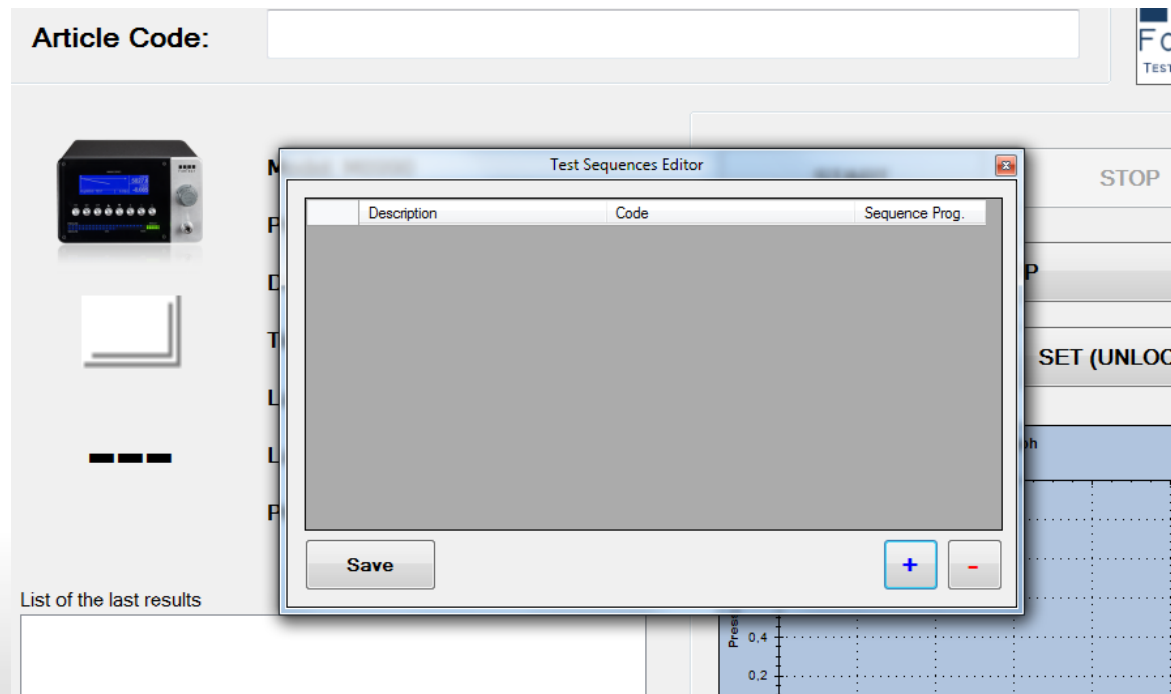
SEQUENCES

Sequences is an editor that permit to execute programs on the device if an article code match with a specific code. For example a supervisor want to perform test of program 1 when the article code is “ABC”, test program 2 when the article code is “123” etc..



SEQUENCES

- + : Add a rule to the sequence editor
- : Remove an existing rule

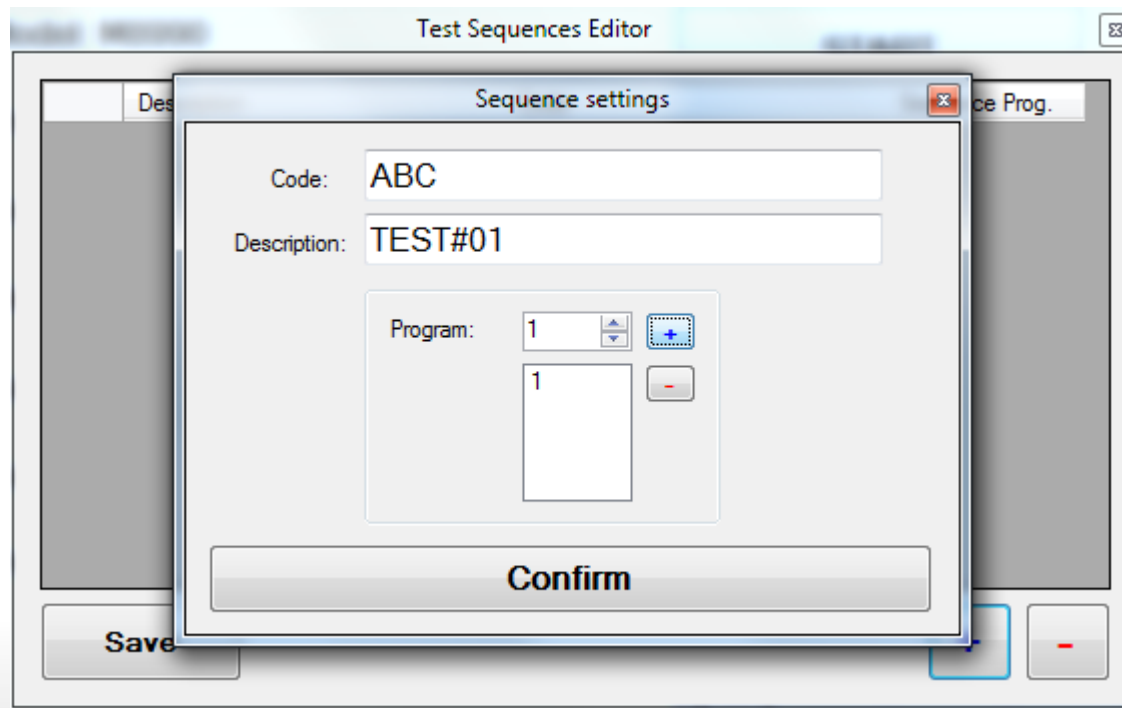


SEQUENCES

Code: is the code to match with the article code

Description: is a description of the test to perform

Program: is the chain of programs to execute



EXCEL ARCHIVE

Example of test archive

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
1	Date	Hour	Program N	Filling Pre	Filling Pre	Filling Tim	Settling Ti	Test Time	Time UM	Max Deca	Decay Ev	Decay UM	VOUT_AL	VOUT_AL	Pressure t	Pressure t	Remaining	Remaining	Last Phase
2	04/11/2014	11:50:32	0	9000	mbar	0,5	0,5	0,2	s	1000	-0,128	mbar	0	0	905,7	mbar	0	s	Fase ad u Bu
3	04/11/2014	11:50:33	0	9000	mbar	0,5	0,5	0,2	s	1000	-0,085	mbar	0	0	907,1	mbar	0	s	Fase ad u Bu
4	04/11/2014	11:50:35	0	9000	mbar	0,5	0,5	0,2	s	1000	-0,073	mbar	0	0	907,2	mbar	0	s	Fase ad u Bu
5	04/11/2014	11:50:38	0	9000	mbar	0,5	0,5	0,2	s	1000	-0,067	mbar	0	0	907,4	mbar	0	s	Fase ad u Bu
6	04/11/2014	11:50:39	0	9000	mbar	0,5	0,5	0,2	s	1000	-0,06	mbar	0	0	907,6	mbar	0	s	Fase ad u Bu
7	04/11/2014	11:50:41	0	9000	mbar	0,5	0,5	0,2	s	1000	-0,06	mbar	0	0	907,5	mbar	0	s	Fase ad u Bu
8	04/11/2014	11:50:43	0	9000	mbar	0,5	0,5	0,2	s	1000	-0,056	mbar	0	0	907,3	mbar	0	s	Fase ad u Bu
9	04/11/2014	11:50:45	0	9000	mbar	0,5	0,5	0,2	s	1000	-0,053	mbar	0	0	907,3	mbar	0	s	Fase ad u Bu
10	04/11/2014	11:50:47	0	9000	mbar	0,5	0,5	0,2	s	1000	-0,051	mbar	0	0	907,9	mbar	0	s	Fase ad u Bu
11	04/11/2014	11:50:49	0	9000	mbar	0,5	0,5	0,2	s	1000	-0,051	mbar	0	0	907,1	mbar	0	s	Fase ad u Bu
12	04/11/2014	11:50:51	0	9000	mbar	0,5	0,5	0,2	s	1000	-0,05	mbar	0	0	906,9	mbar	0	s	Fase ad u Bu
13	04/11/2014	11:50:53	0	9000	mbar	0,5	0,5	0,2	s	1000	-0,046	mbar	0	0	907,6	mbar	0	s	Fase ad u Bu
14	04/11/2014	11:50:55	0	9000	mbar	0,5	0,5	0,2	s	1000	-0,047	mbar	0	0	907,4	mbar	0	s	Fase ad u Bu
15	04/11/2014	11:50:57	0	9000	mbar	0,5	0,5	0,2	s	1000	-0,052	mbar	0	0	906,2	mbar	0	s	Fase ad u Bu